

Import Libraries

```
In [1]: import numpy as np
import pandas as pd
import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

```
In [2]: import seaborn as sns
import matplotlib.pyplot as plt
import scipy.stats as st
%matplotlib inline

sns.set(style="whitegrid")
```

```
In [3]: import warnings
warnings.filterwarnings('ignore')
```

Import dataset

```
In [10]: df = pd.read_csv(r"C:\Users\suras\OneDrive\NARESH IT\Practice tasks\heart.csv")
```

```
In [11]: print('The shape of the dataset : ', df.shape)
```

The shape of the dataset : (303, 14)

```
In [12]: df.head()
```

Out[12]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	ti
--	-----	-----	----	----------	------	-----	---------	---------	-------	---------	-------	----	------	----

0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	



```
In [13]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   age         303 non-null   int64
 1   sex         303 non-null   int64
 2   cp          303 non-null   int64
 3   trestbps    303 non-null   int64
 4   chol        303 non-null   int64
 5   fbs         303 non-null   int64
 6   restecg     303 non-null   int64
 7   thalach     303 non-null   int64
 8   exang       303 non-null   int64
 9   oldpeak     303 non-null   float64
10   slope       303 non-null   int64
11   ca          303 non-null   int64
12   thal        303 non-null   int64
13   target      303 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

```
In [14]: df.dtypes
```

```
Out[14]: age          int64
sex            int64
cp             int64
trestbps       int64
chol           int64
fbs            int64
restecg        int64
thalach        int64
exang          int64
oldpeak        float64
slope          int64
ca             int64
thal           int64
target         int64
dtype: object
```

```
In [15]: df.describe()
```

Out[15]:

	age	sex	cp	trestbps	chol	fbs	restecg
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000

In [16]: `df.columns`

Out[16]: Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'], dtype='object')

In [17]: `df['target'].nunique()`

Out[17]: 2

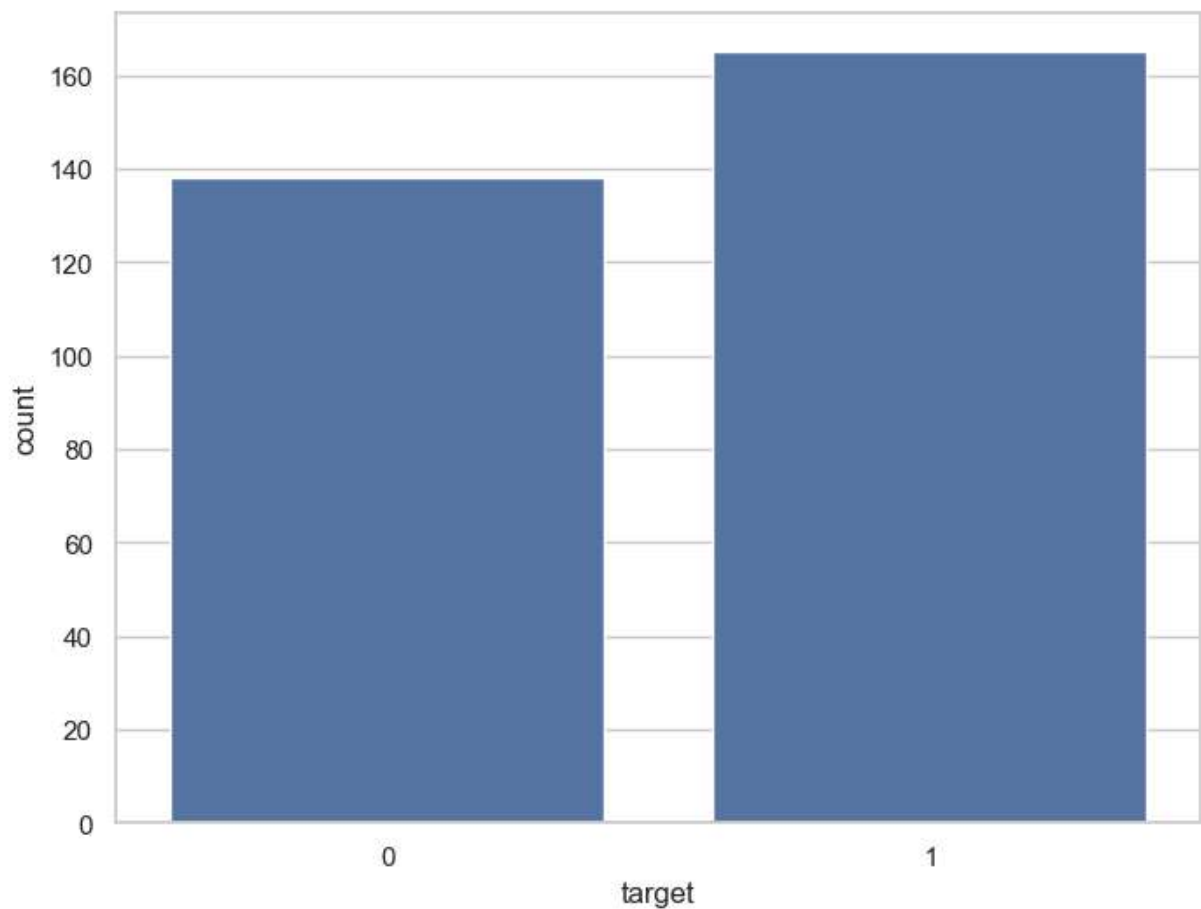
In [18]: `df['target'].unique()`

Out[18]: array([1, 0], dtype=int64)

In [19]: `df['target'].value_counts()`

Out[19]: target
 1 165
 0 138
 Name: count, dtype: int64

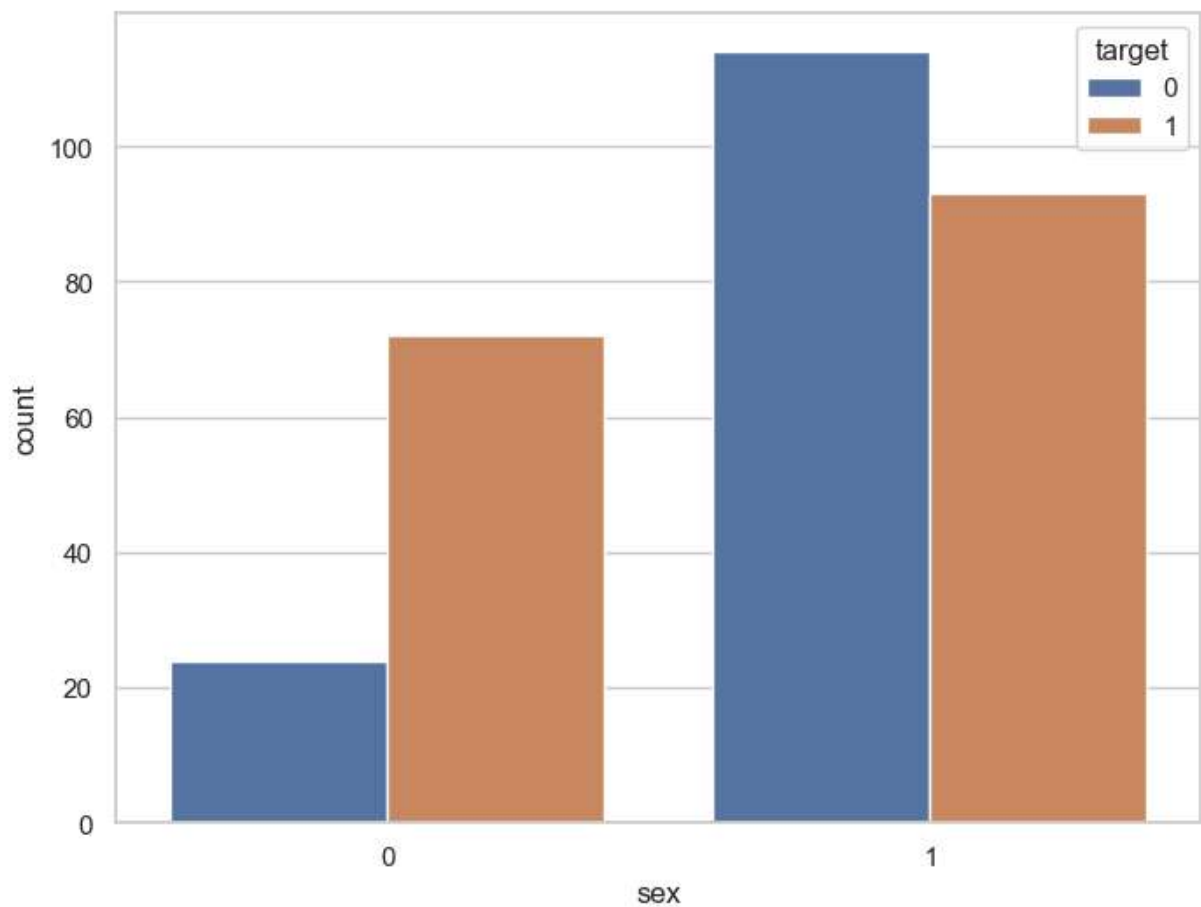
In [21]: `f, ax = plt.subplots(figsize=(8, 6))
 ax = sns.countplot(x="target", data=df)
 plt.show()`



```
In [22]: df.groupby('sex')['target'].value_counts()
```

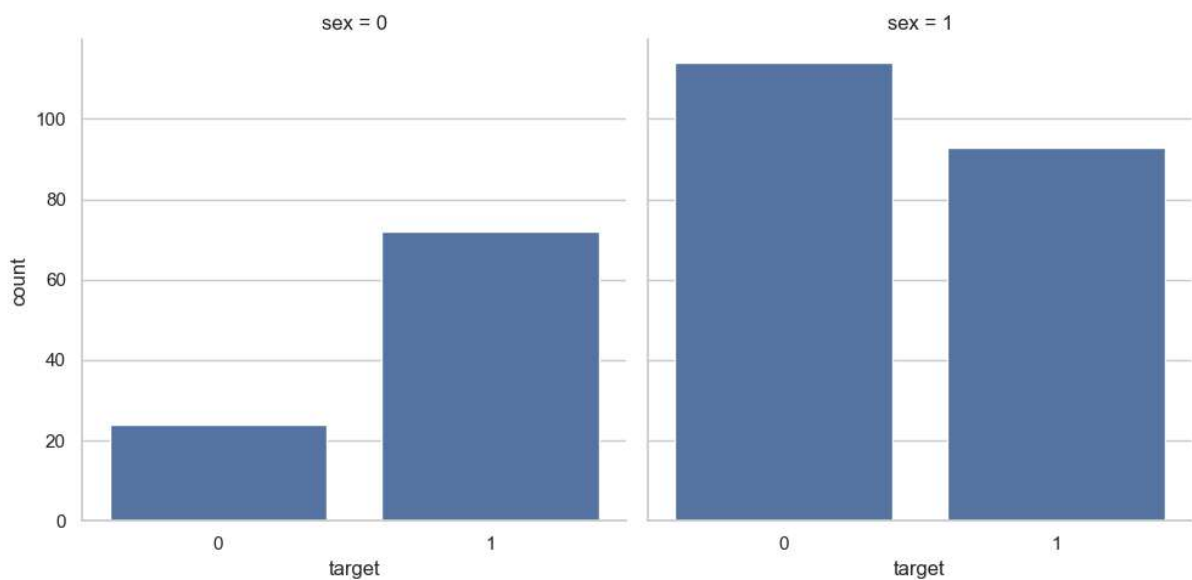
```
Out[22]: sex  target
0      1         72
      0         24
1      0        114
      1         93
Name: count, dtype: int64
```

```
In [23]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="sex", hue="target", data=df)
plt.show()
```

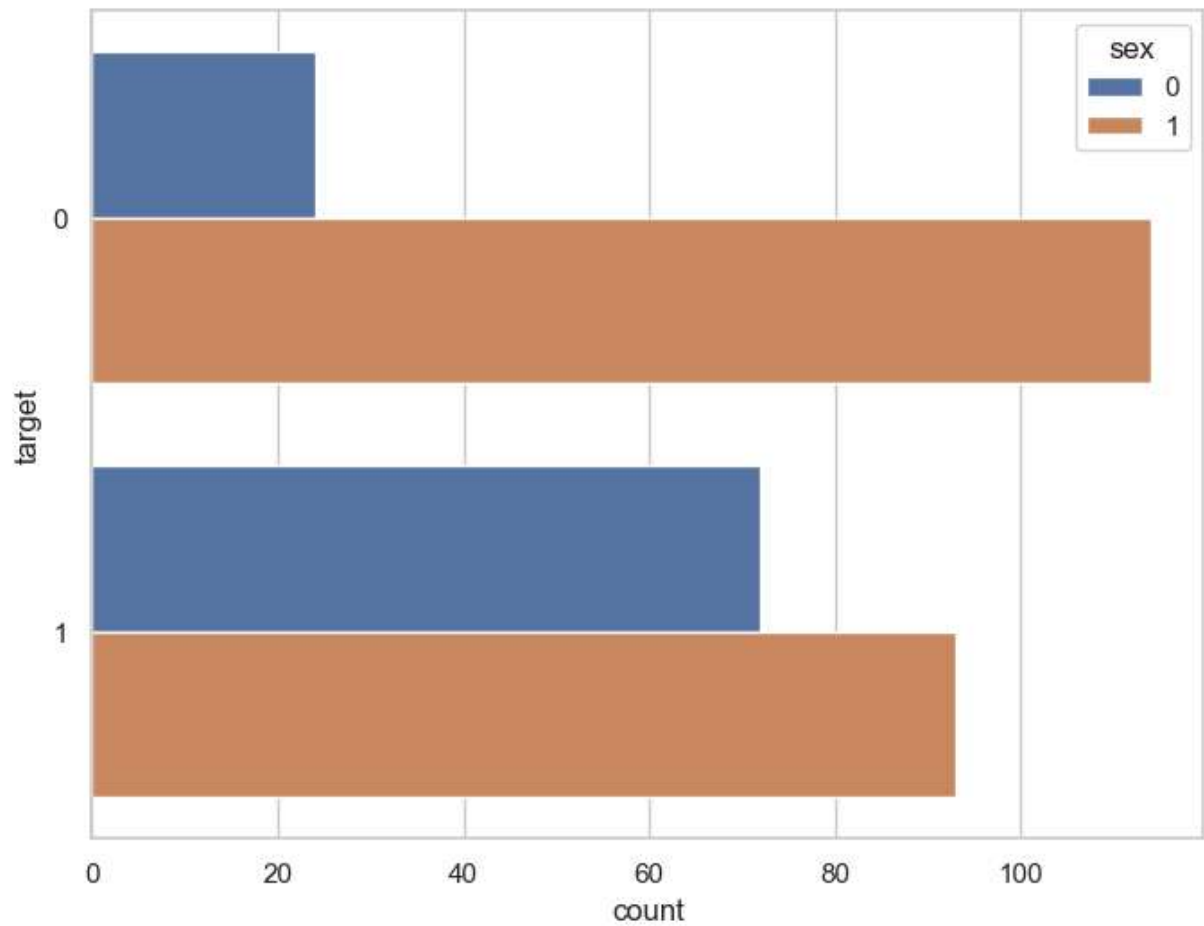


```
In [26]: ax = sns.catplot(x="target", col="sex", data=df, kind="count", height=5, aspect=1)
```

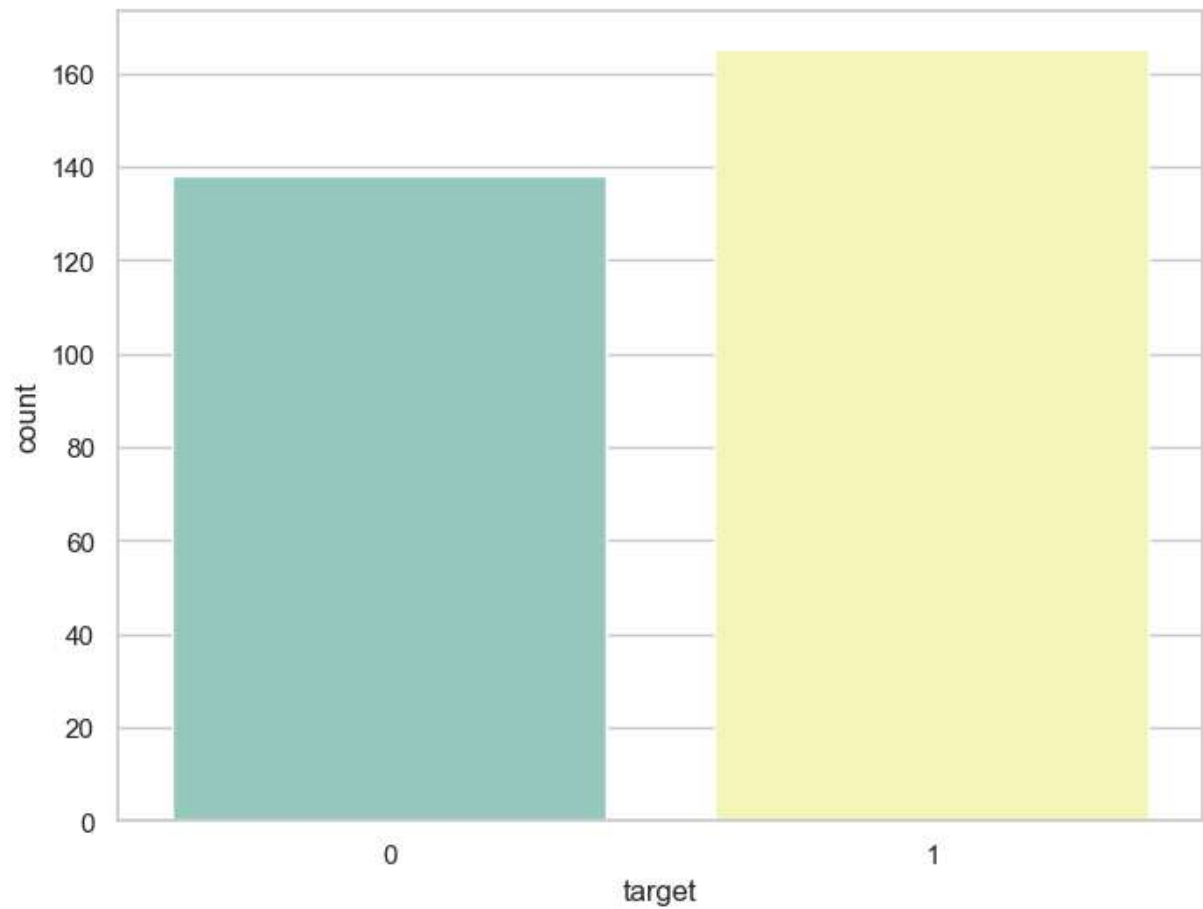
```
In [27]: plt.show()
```



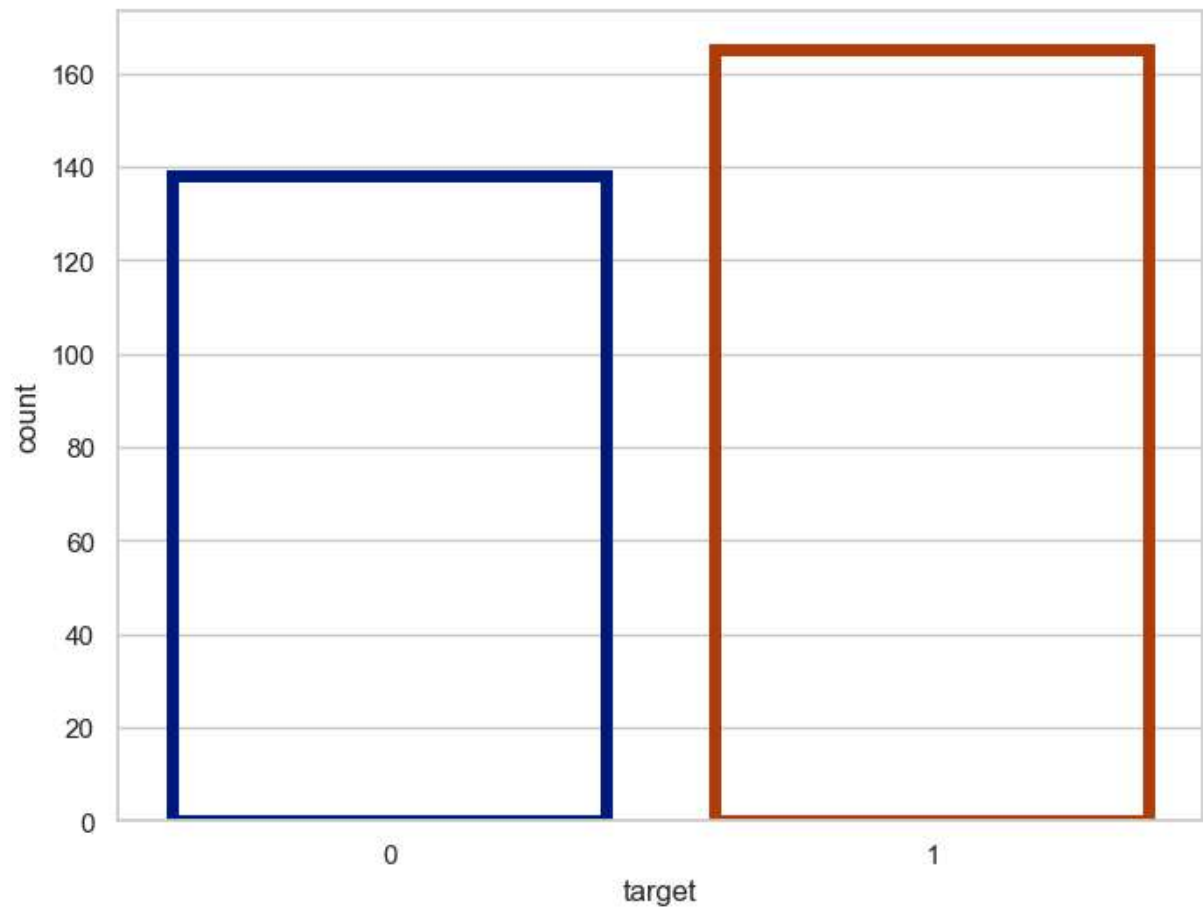
```
In [28]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(y="target", hue="sex", data=df)  
plt.show()
```



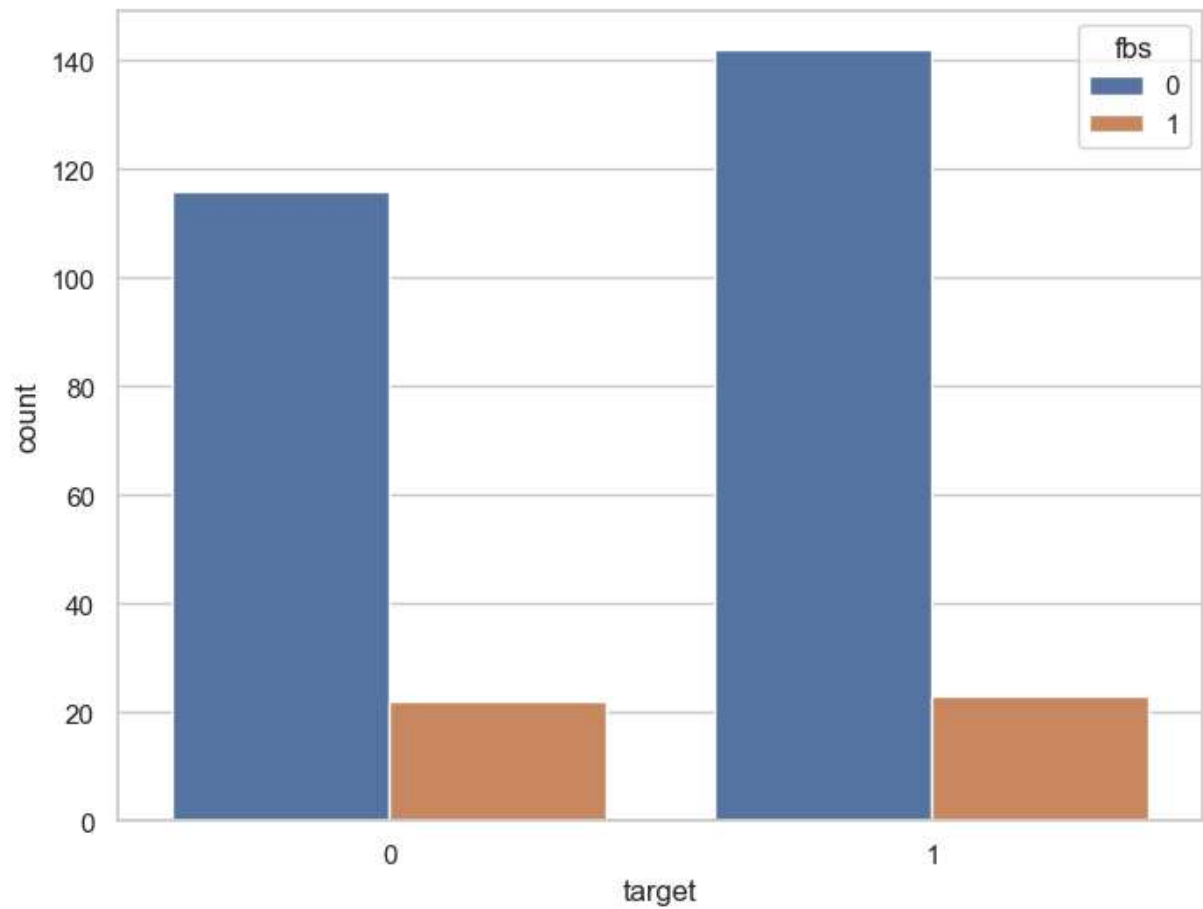
```
In [29]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", data=df, palette="Set3")
plt.show()
```



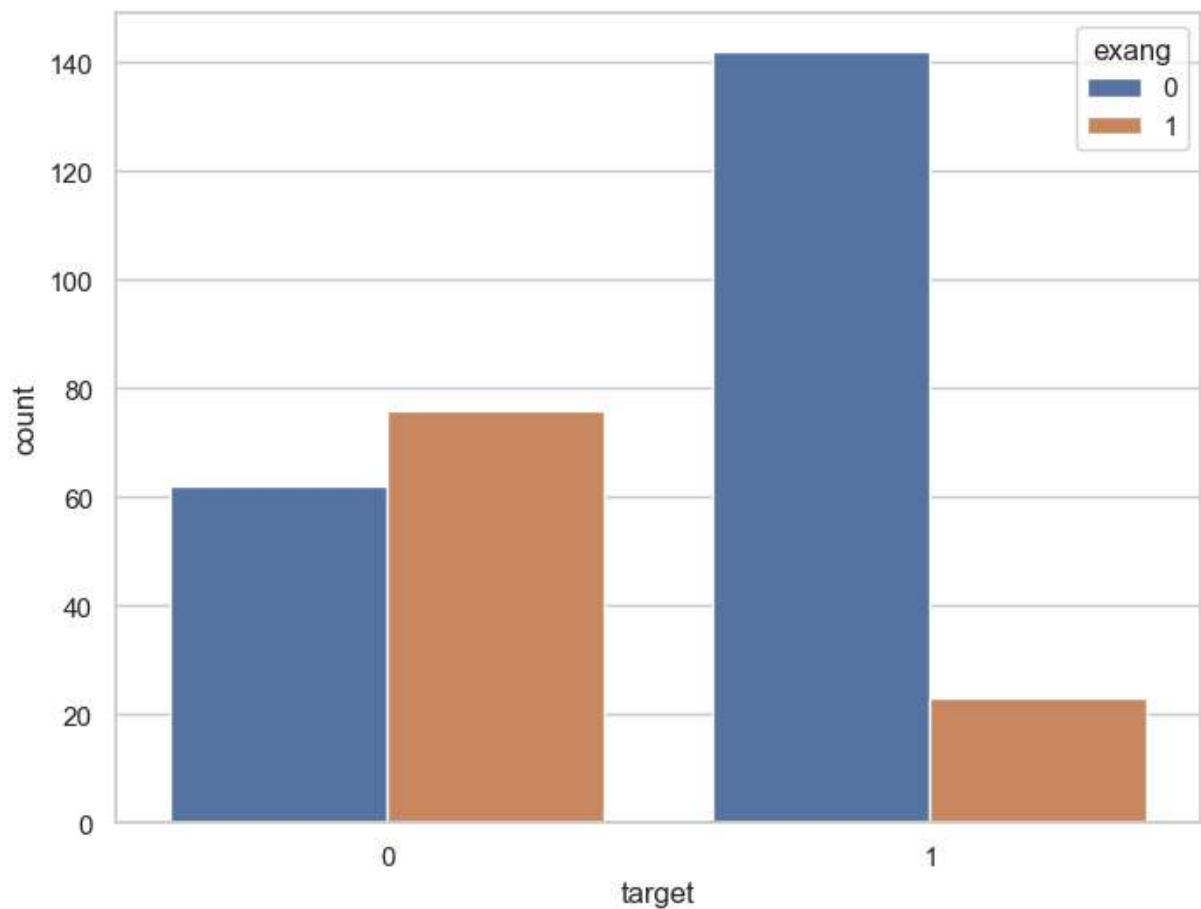
```
In [30]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", data=df, facecolor=(0, 0, 0, 0), linewidth=5, edgeco
plt.show()
```



```
In [31]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(x="target", hue="fbs", data=df)  
plt.show()
```

```
In [32]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", hue="fbs", data=df)
plt.show()
```



```
In [33]: correlation = df.corr()
```

```
In [34]: correlation['target'].sort_values(ascending=False)
```

```
Out[34]: target      1.000000
         cp         0.433798
         thalach    0.421741
         slope      0.345877
         restecg     0.137230
         fbs        -0.028046
         chol       -0.085239
         trestbps   -0.144931
         age        -0.225439
         sex        -0.280937
         thal       -0.344029
         ca         -0.391724
         oldpeak    -0.430696
         exang      -0.436757
         Name: target, dtype: float64
```

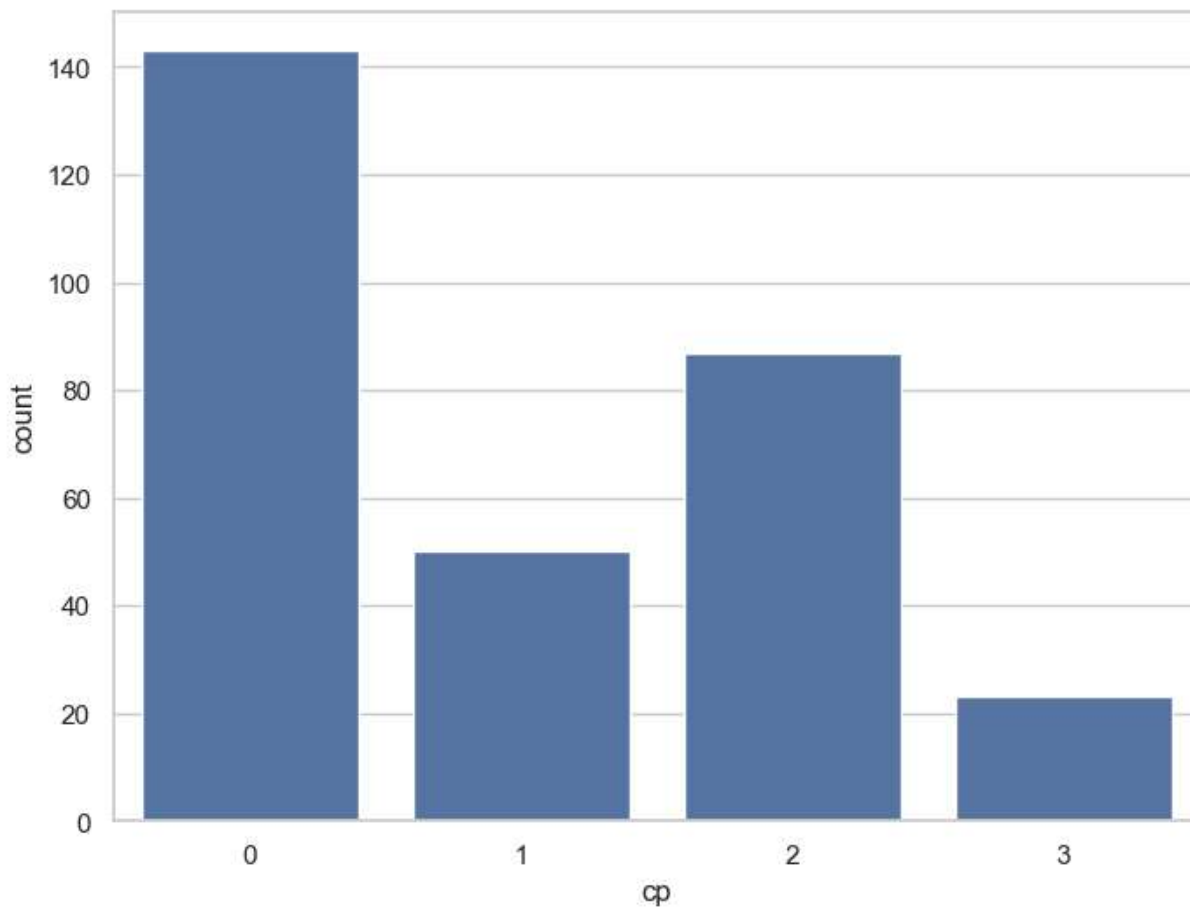
```
In [35]: df['cp'].nunique()
```

```
Out[35]: 4
```

```
In [36]: df['cp'].value_counts()
```

```
Out[36]: cp
0      143
2       87
1       50
3       23
Name: count, dtype: int64
```

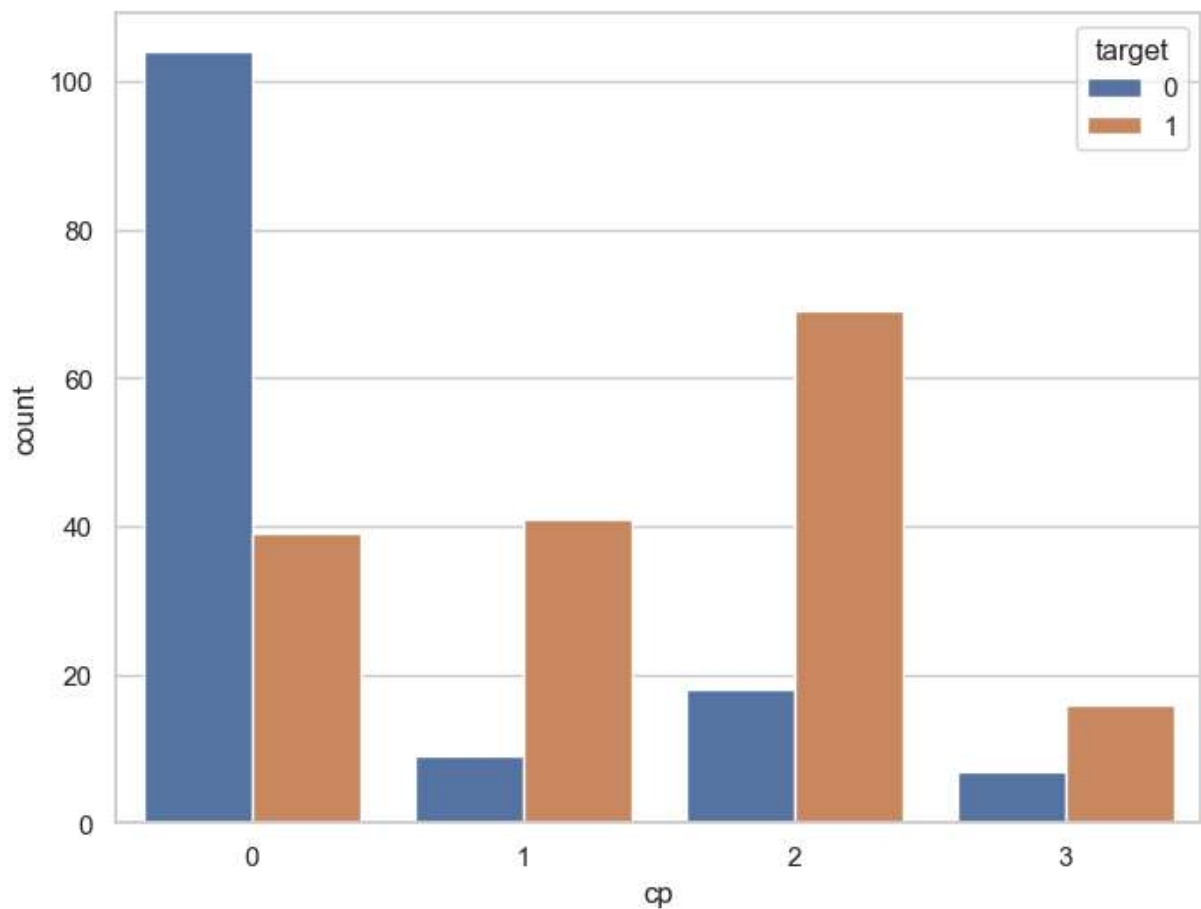
```
In [38]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="cp", data=df)
plt.show()
```



```
In [39]: df.groupby('cp')['target'].value_counts()
```

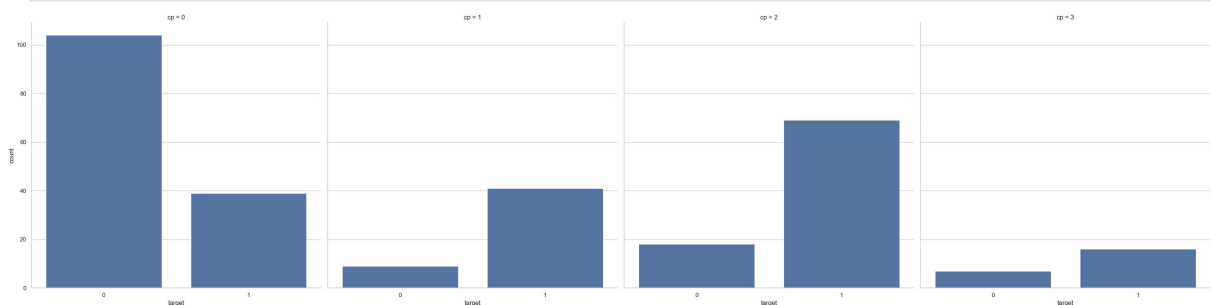
```
Out[39]: cp  target
0      0      104
        1       39
1      1       41
        0        9
2      1       69
        0       18
3      1       16
        0        7
Name: count, dtype: int64
```

```
In [40]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="cp", hue="target", data=df)
plt.show()
```



```
In [41]: ax = sns.catplot(x="target", col="cp", data=df, kind="count", height=8, aspect=1)
```

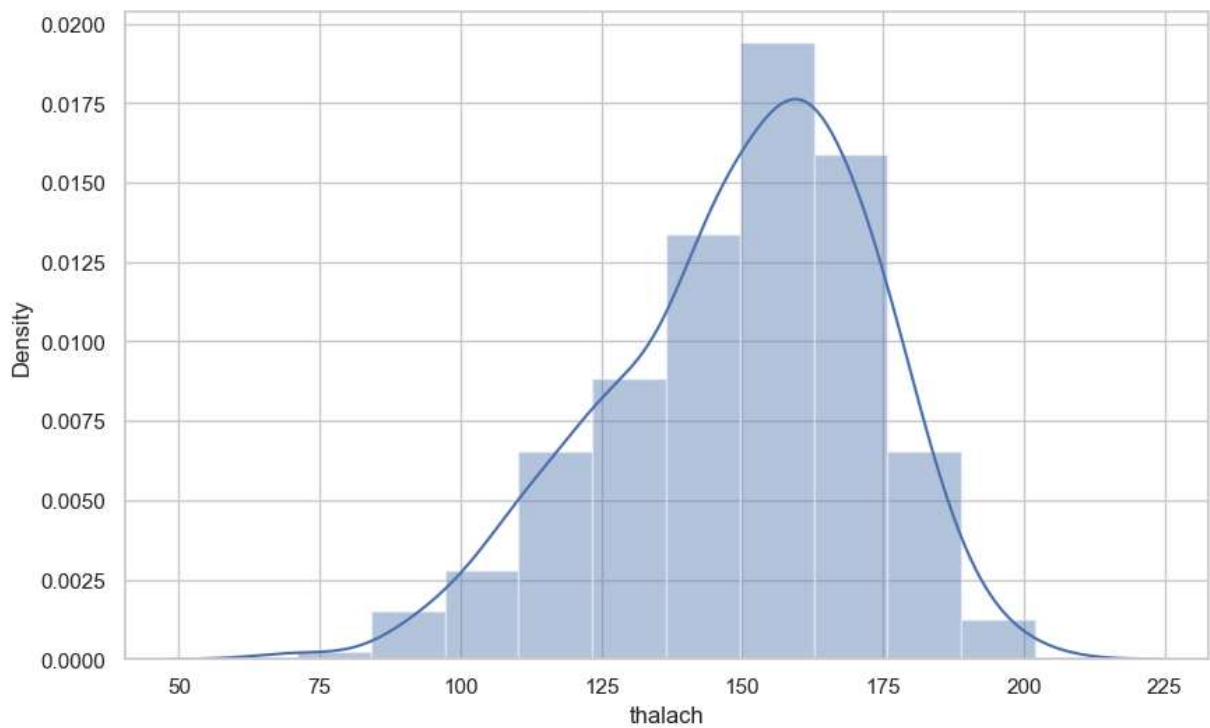
```
In [42]: plt.show()
```



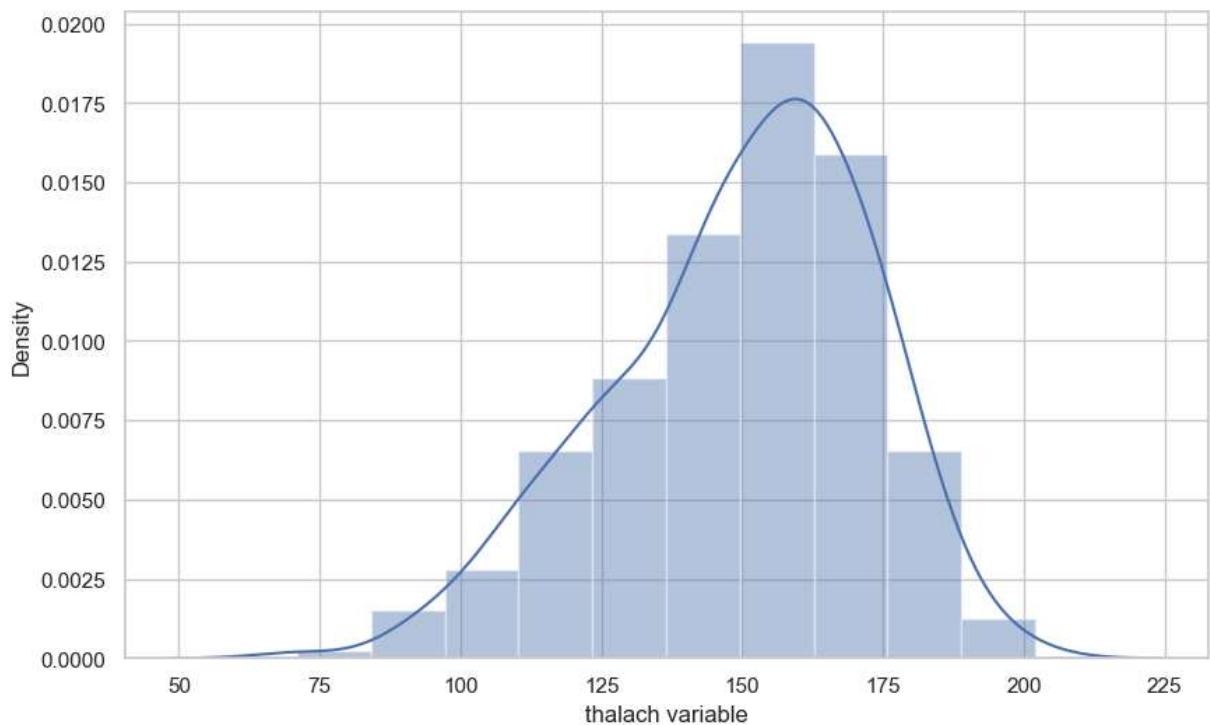
```
In [43]: df['thalach'].nunique()
```

```
Out[43]: 91
```

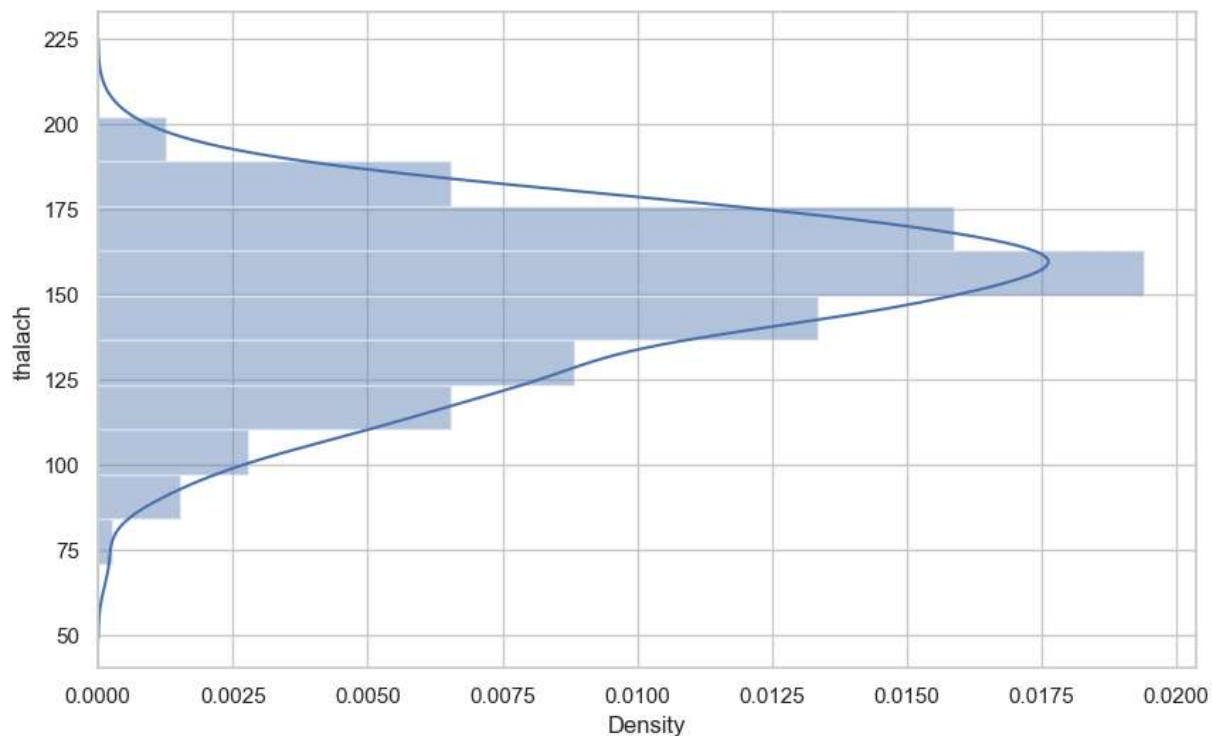
```
In [44]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, bins=10)
plt.show()
```



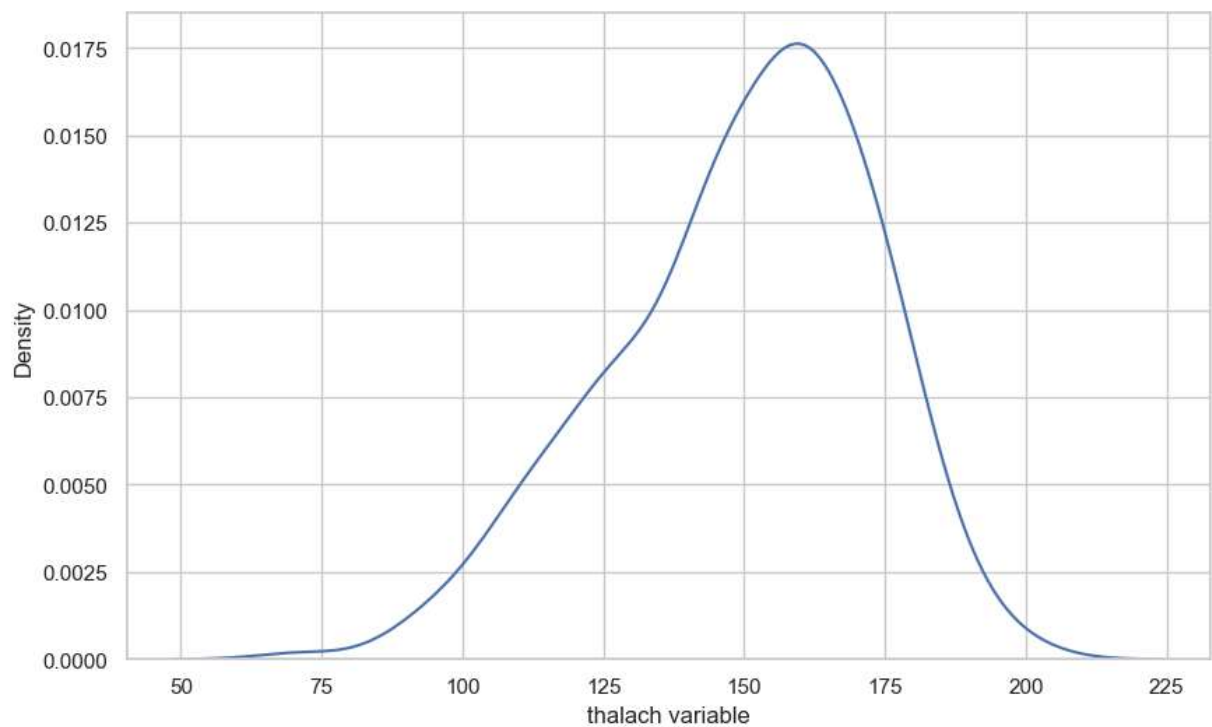
```
In [45]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.distplot(x, bins=10)
plt.show()
```



```
In [46]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, bins=10, vertical=True)
plt.show()
```

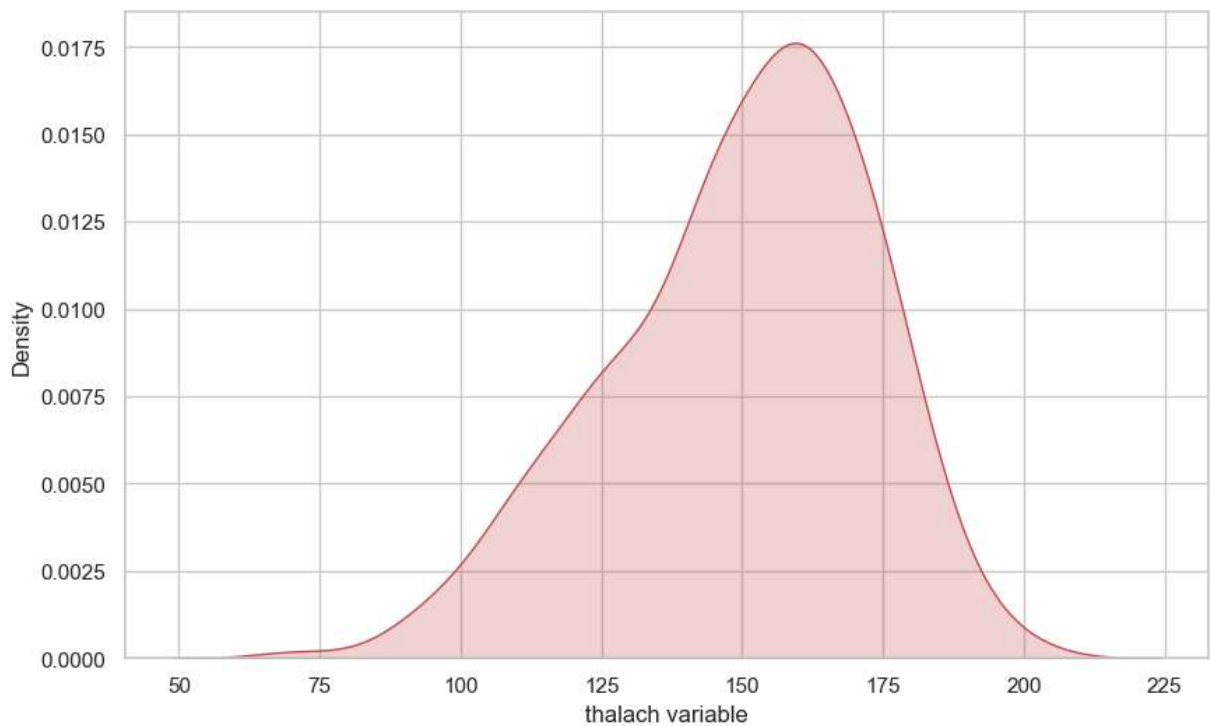


```
In [47]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.kdeplot(x)
plt.show()
```

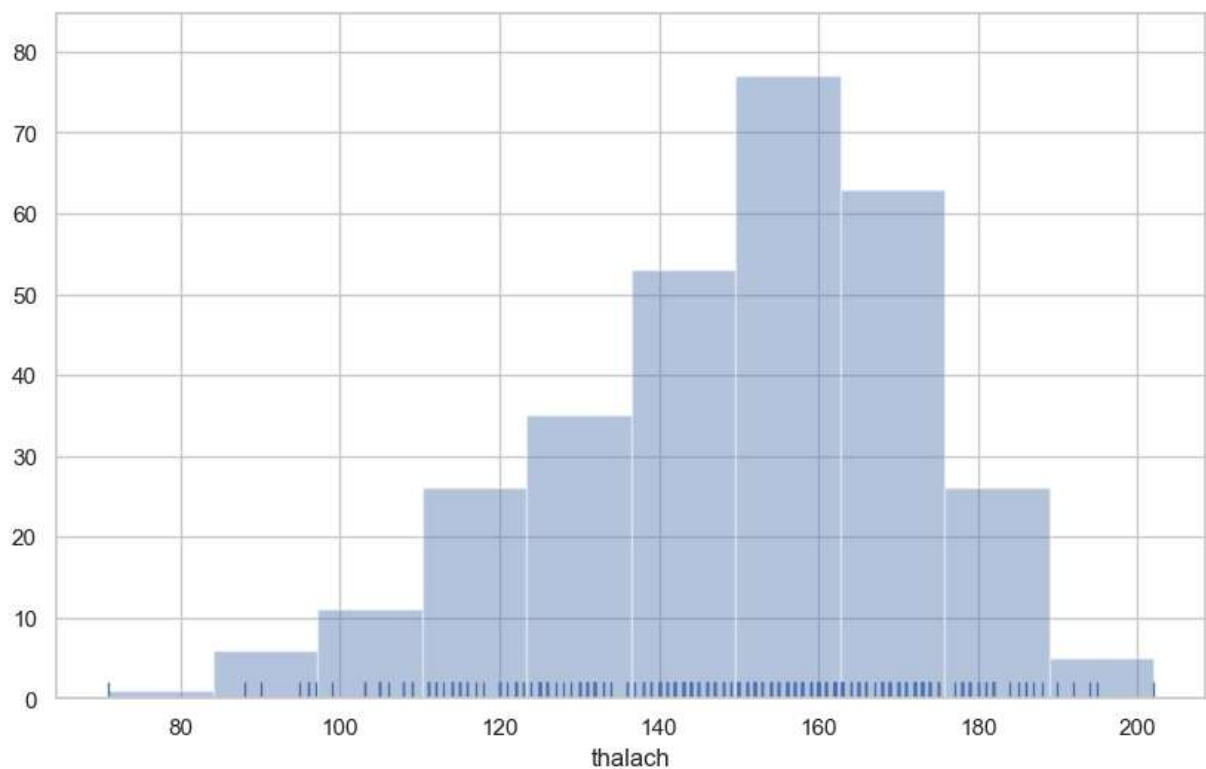


```
In [48]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
```

```
ax = sns.kdeplot(x, shade=True, color='r')  
plt.show()
```

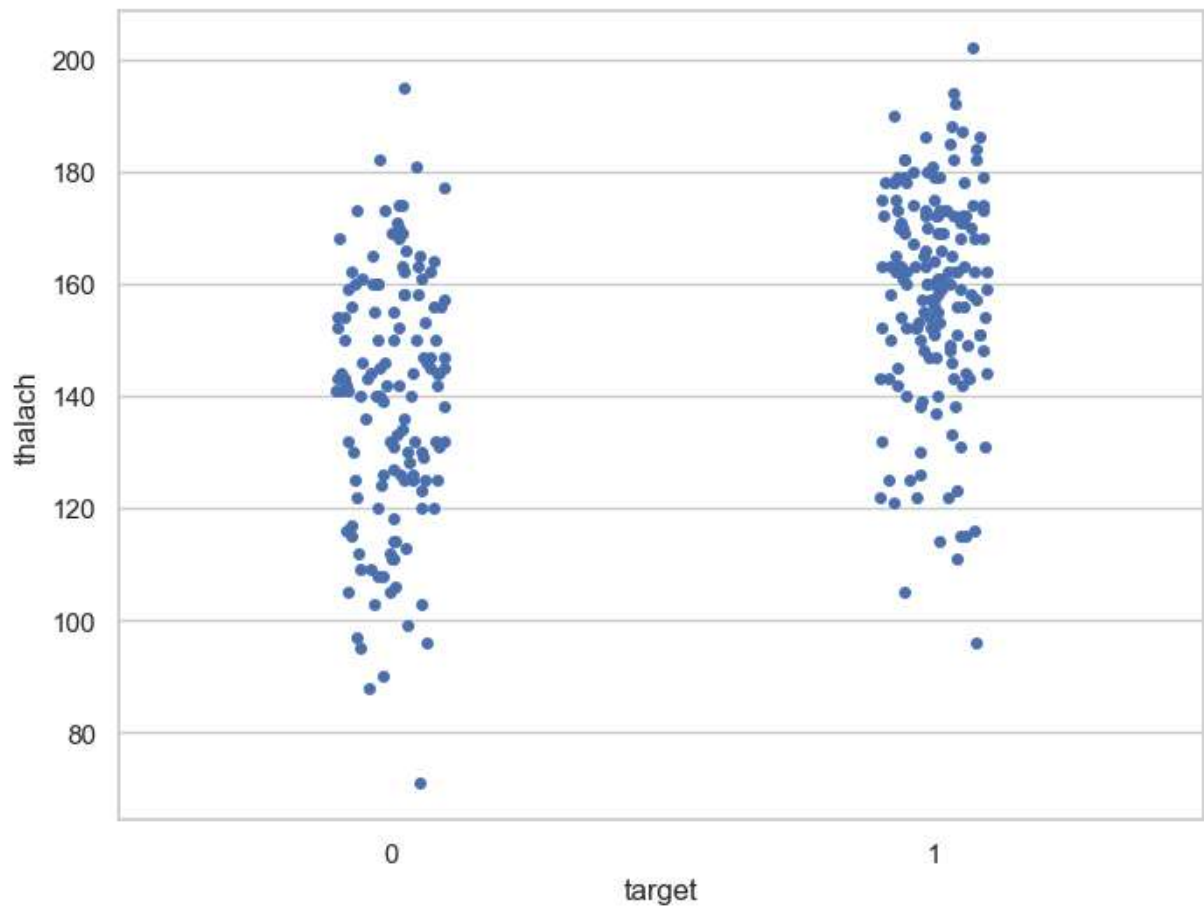


```
In [49]: f, ax = plt.subplots(figsize=(10,6))  
x = df['thalach']  
ax = sns.distplot(x, kde=False, rug=True, bins=10)  
plt.show()
```

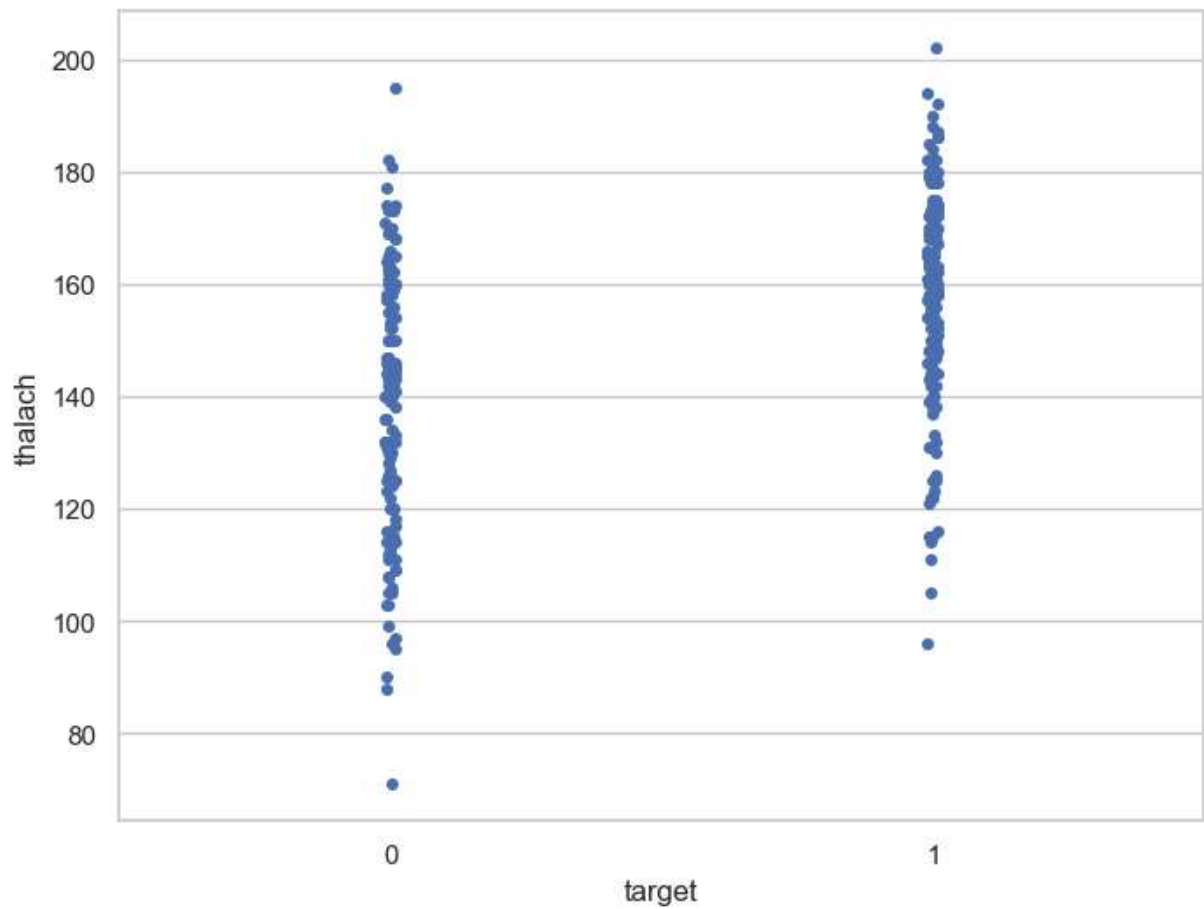


```
In [50]: f, ax = plt.subplots(figsize=(8, 6))  
sns.stripplot(x="target", y="thalach", data=df)
```

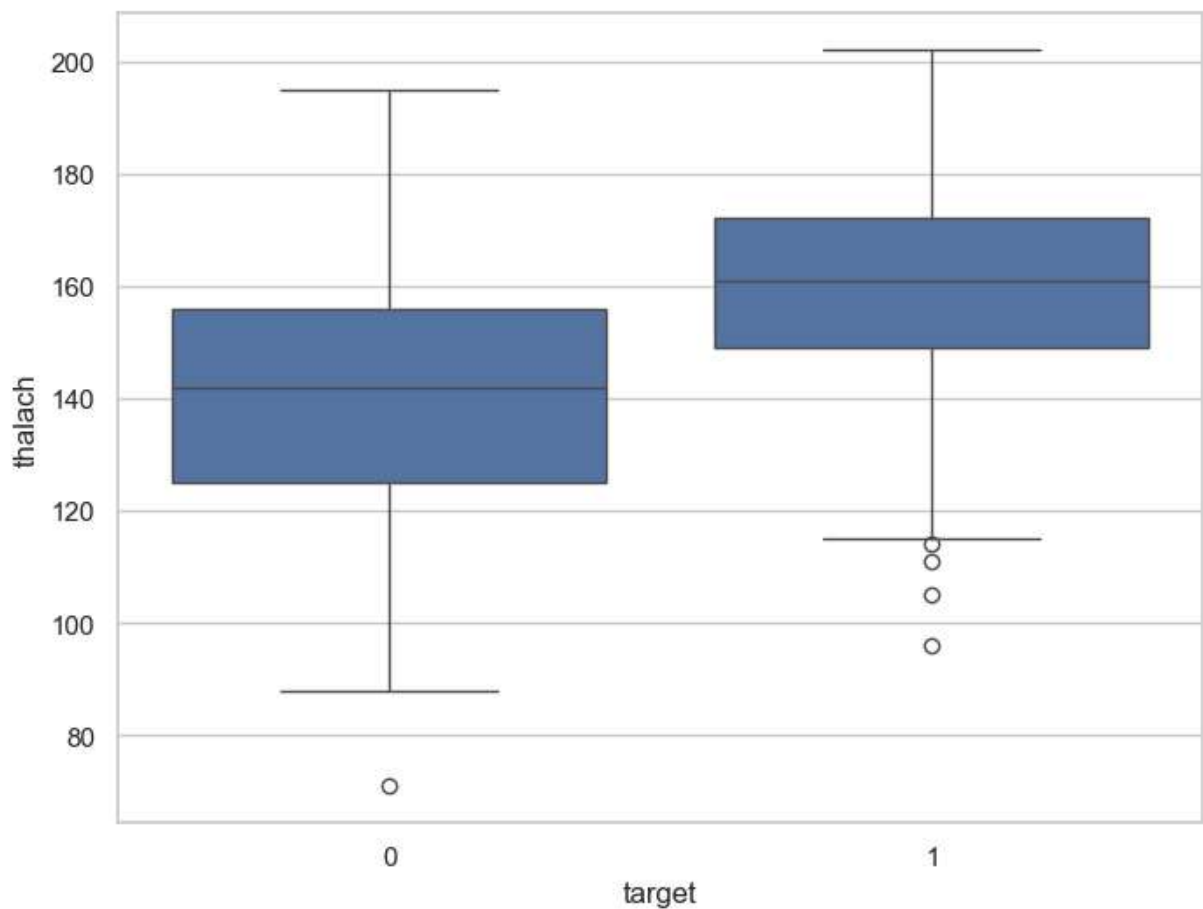
```
plt.show()
```



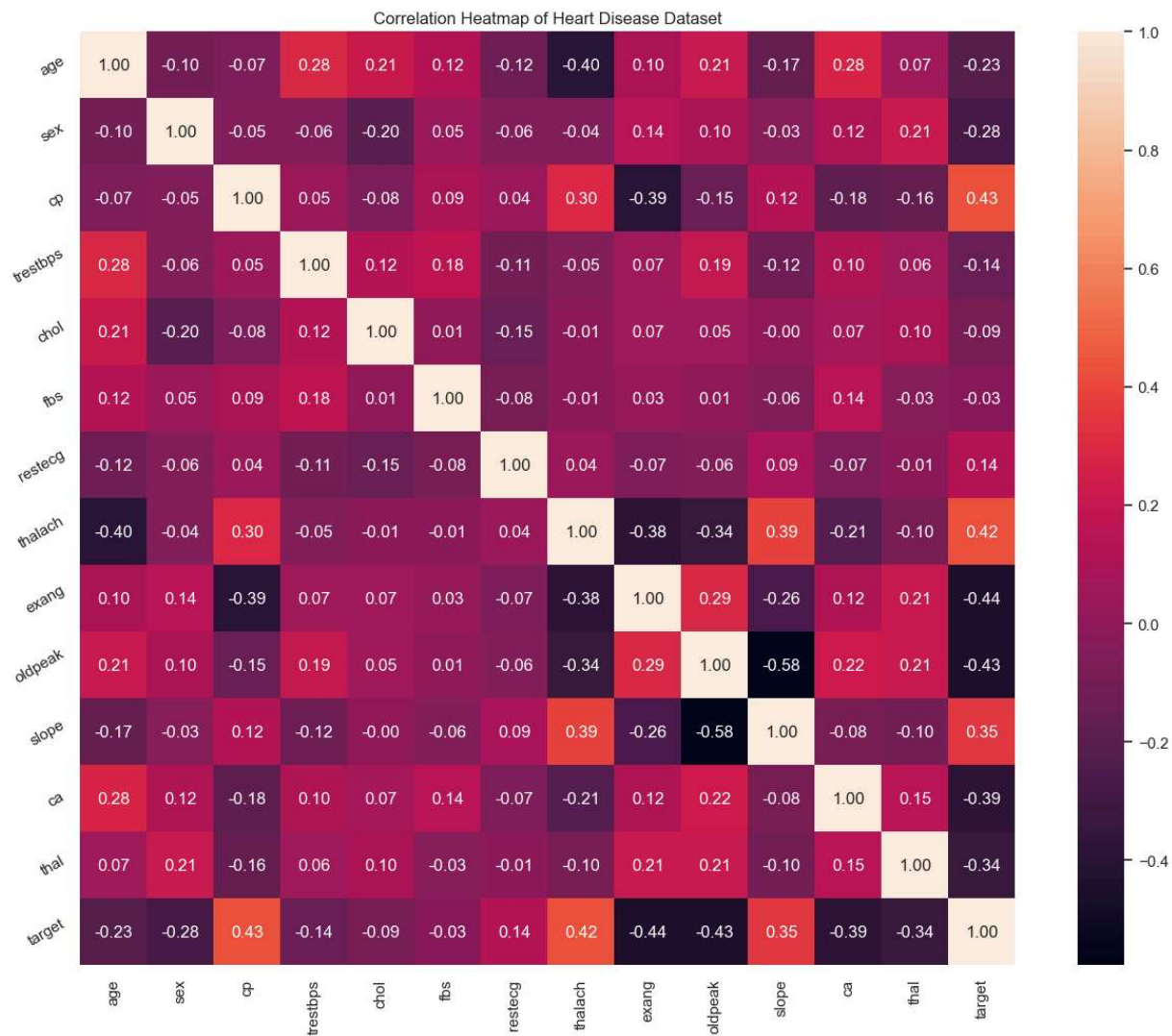
```
In [51]: f, ax = plt.subplots(figsize=(8, 6))  
sns.stripplot(x="target", y="thalach", data=df, jitter = 0.01)  
plt.show()
```

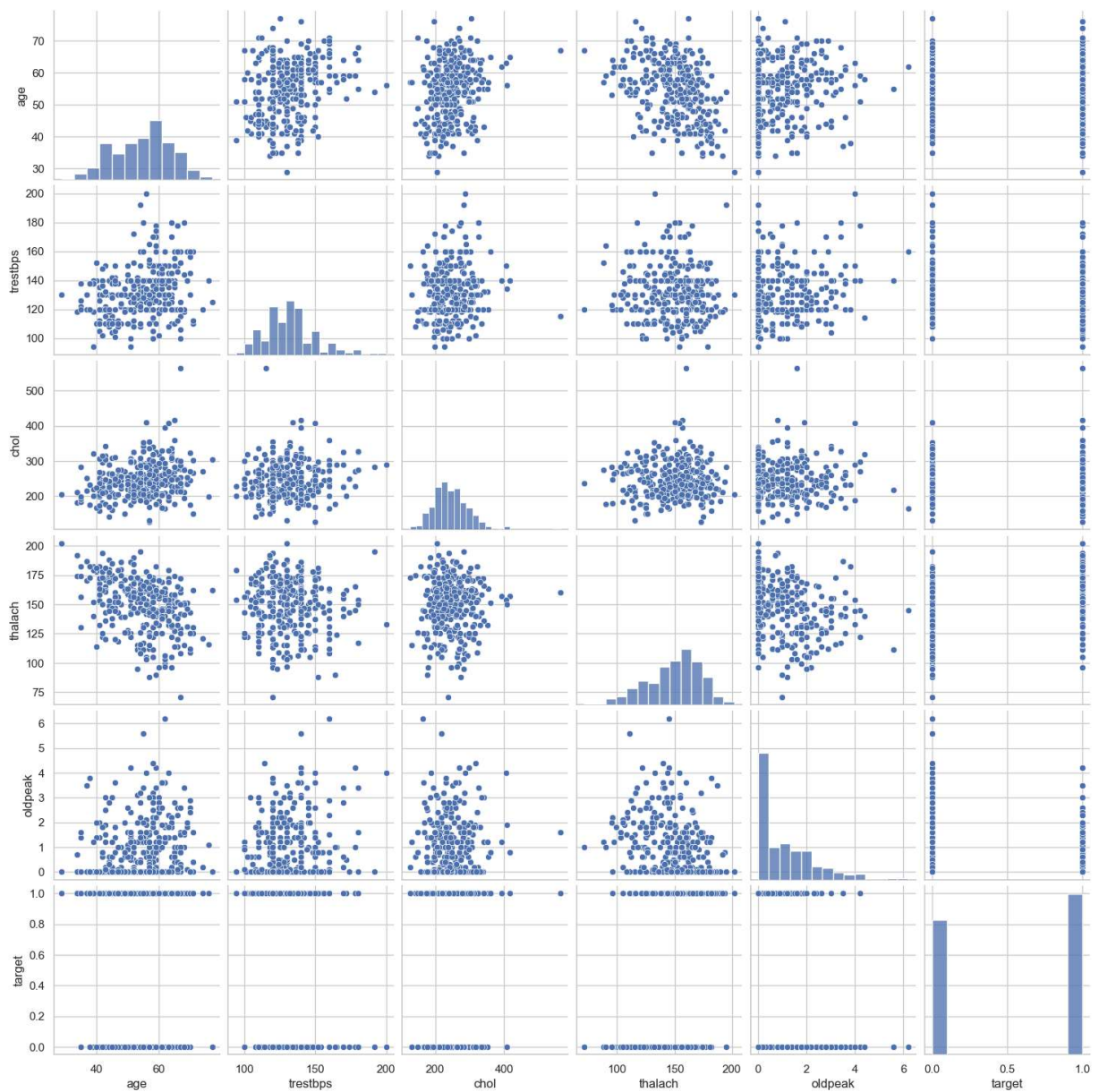
```
In [52]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x="target", y="thalach", data=df)  
plt.show()
```



```
In [53]: plt.figure(figsize=(16,12))
plt.title('Correlation Heatmap of Heart Disease Dataset')
a = sns.heatmap(correlation, square=True, annot=True, fmt='.2f', linecolor='white')
a.set_xticklabels(a.get_xticklabels(), rotation=90)
a.set_yticklabels(a.get_yticklabels(), rotation=30)
plt.show()
```



```
In [54]: num_var = ['age', 'trestbps', 'chol', 'thalach', 'oldpeak', 'target']
sns.pairplot(df[num_var], kind='scatter', diag_kind='hist')
plt.show()
```



```
In [55]: df['age'].nunique()
```

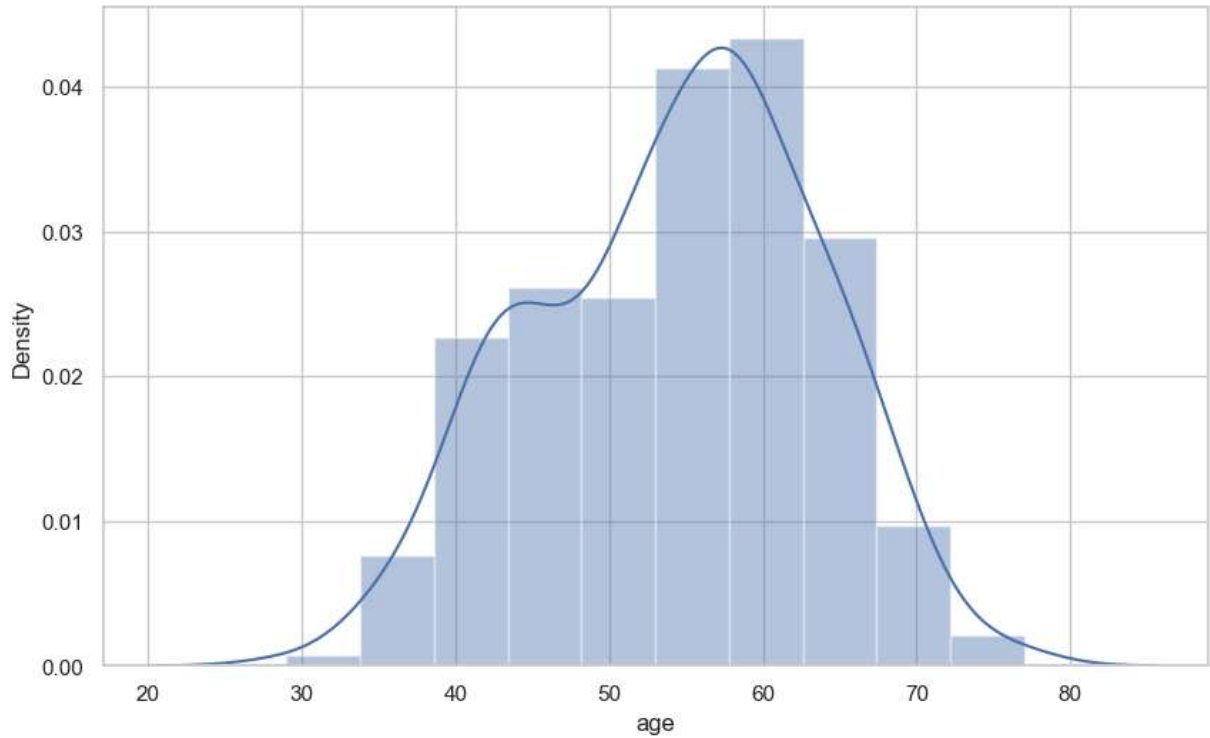
```
Out[55]: 41
```

```
In [56]: df['age'].describe()
```

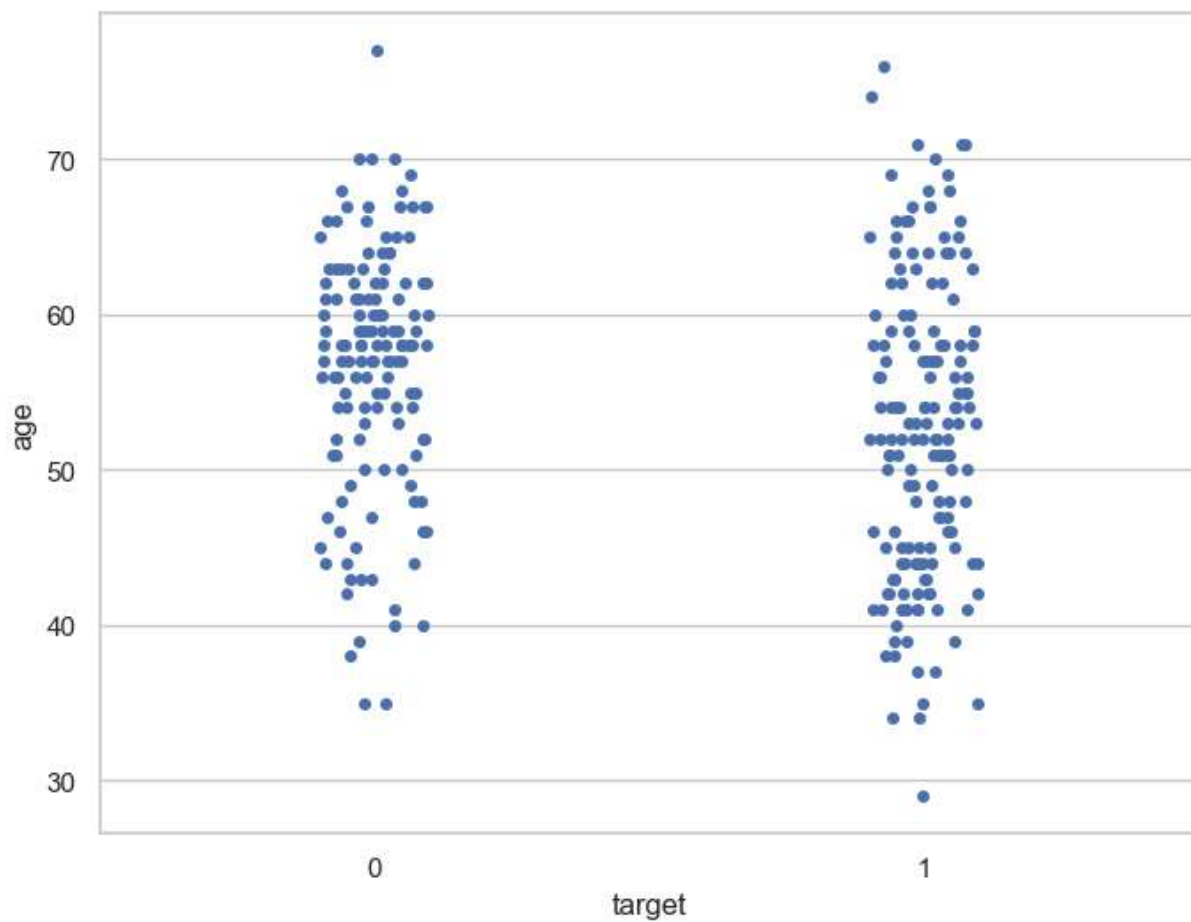
```
Out[56]: count    303.000000
mean       54.366337
std        9.082101
min        29.000000
25%        47.500000
50%        55.000000
75%        61.000000
max        77.000000
Name: age, dtype: float64
```

```
In [57]: f, ax = plt.subplots(figsize=(10,6))
x = df['age']
```

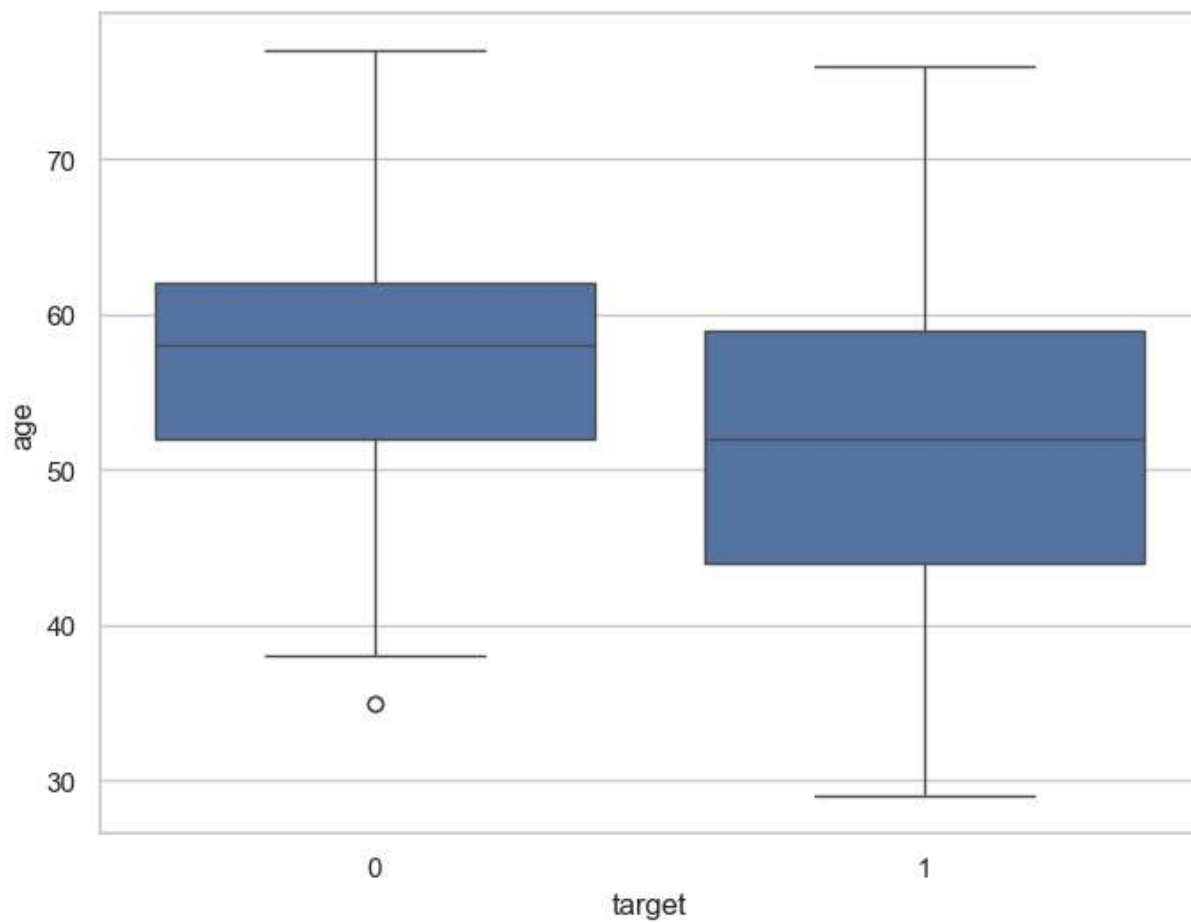
```
ax = sns.distplot(x, bins=10)  
plt.show()
```



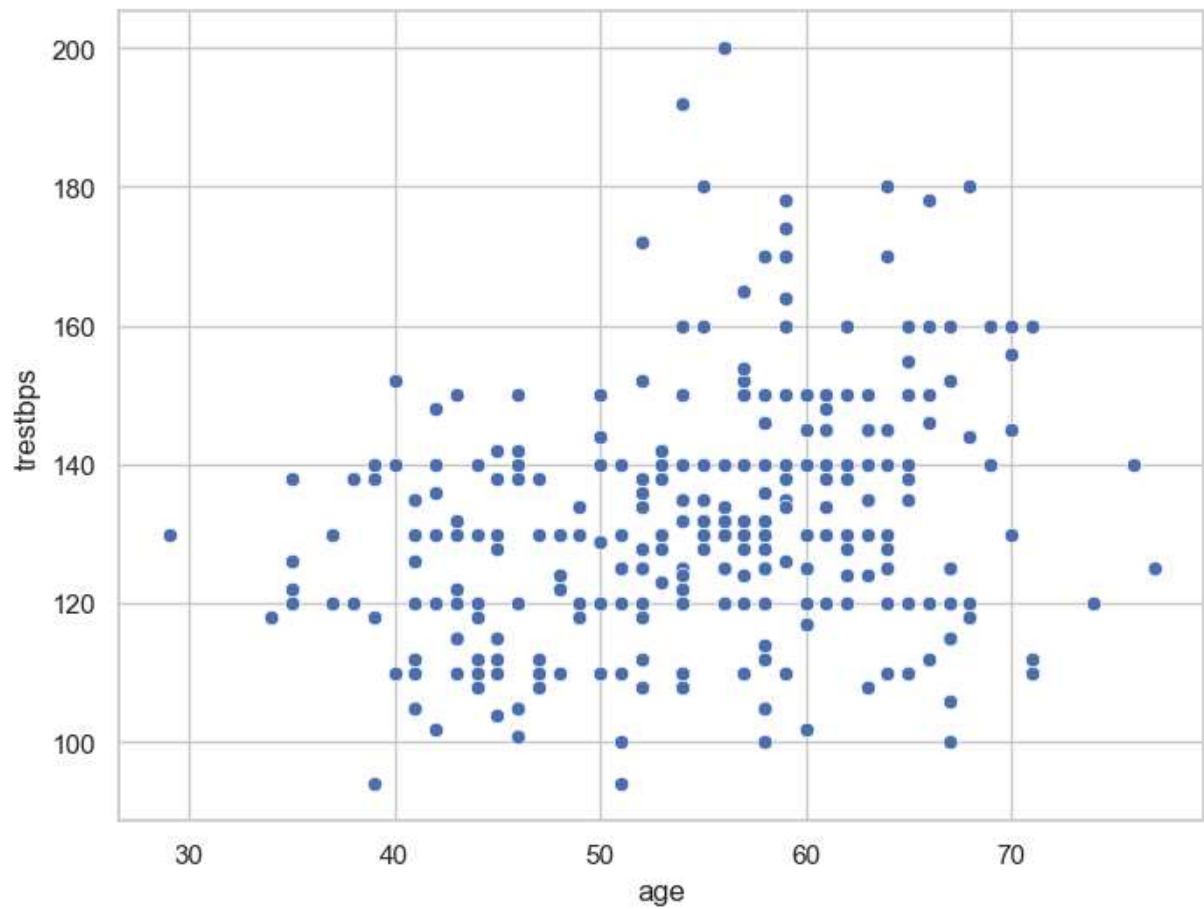
```
In [58]: f, ax = plt.subplots(figsize=(8, 6))  
sns.stripplot(x="target", y="age", data=df)  
plt.show()
```



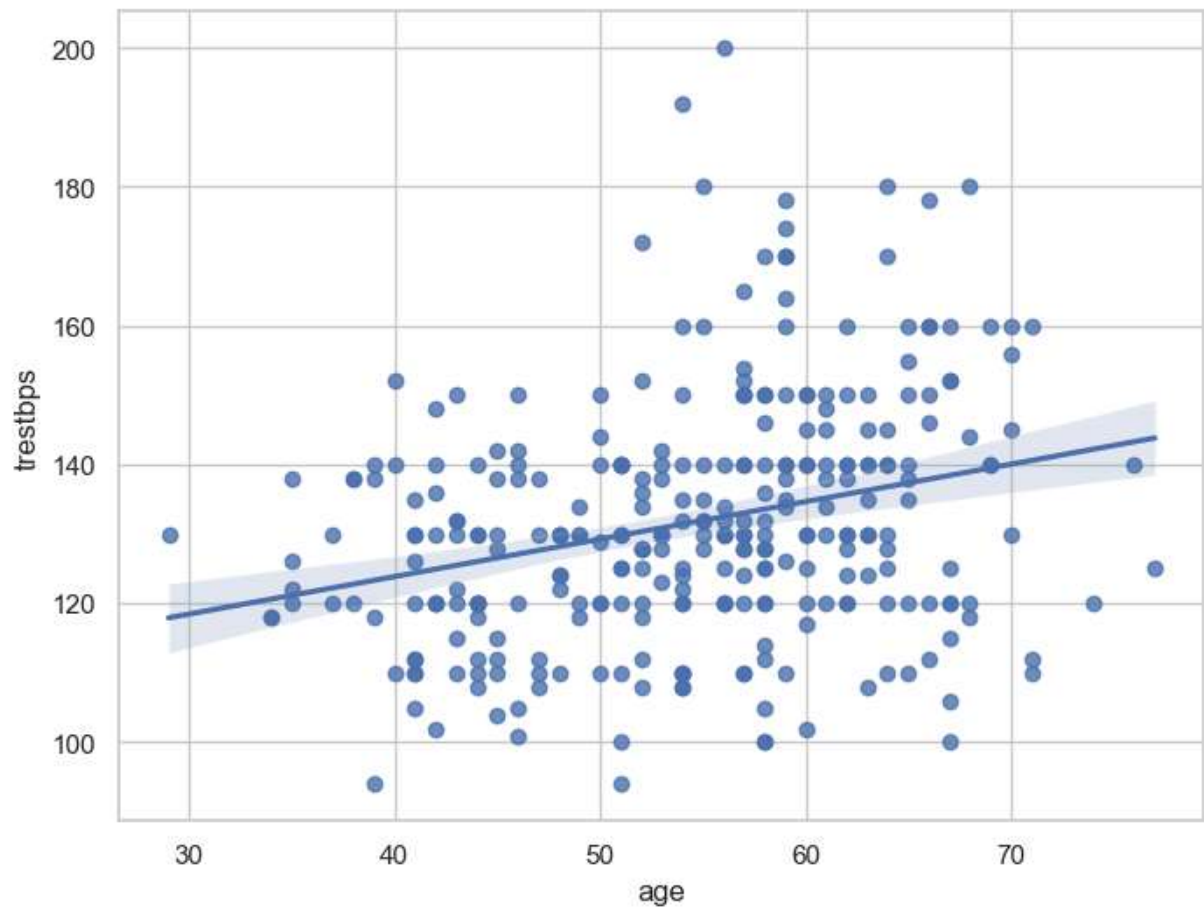
```
In [59]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x="target", y="age", data=df)  
plt.show()
```



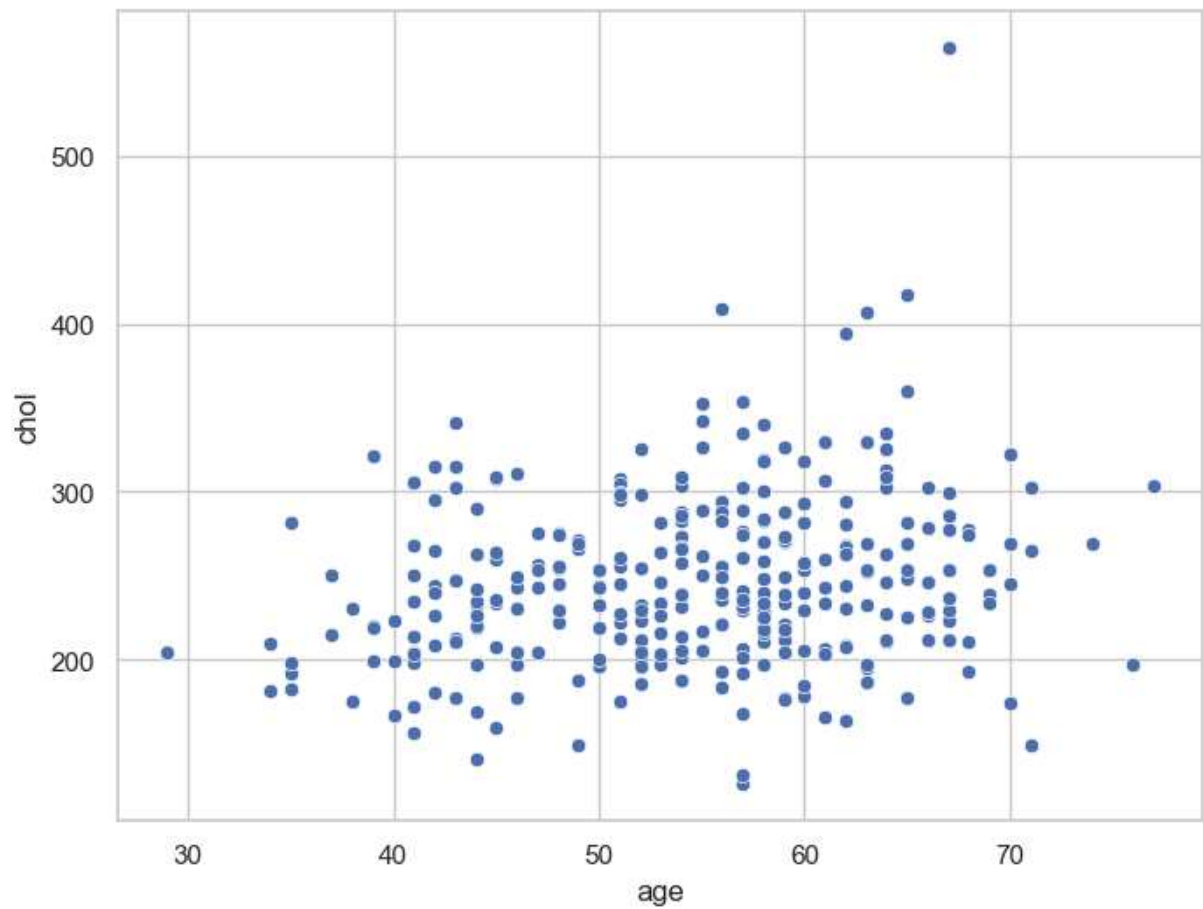
```
In [60]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.scatterplot(x="age", y="trestbps", data=df)  
plt.show()
```



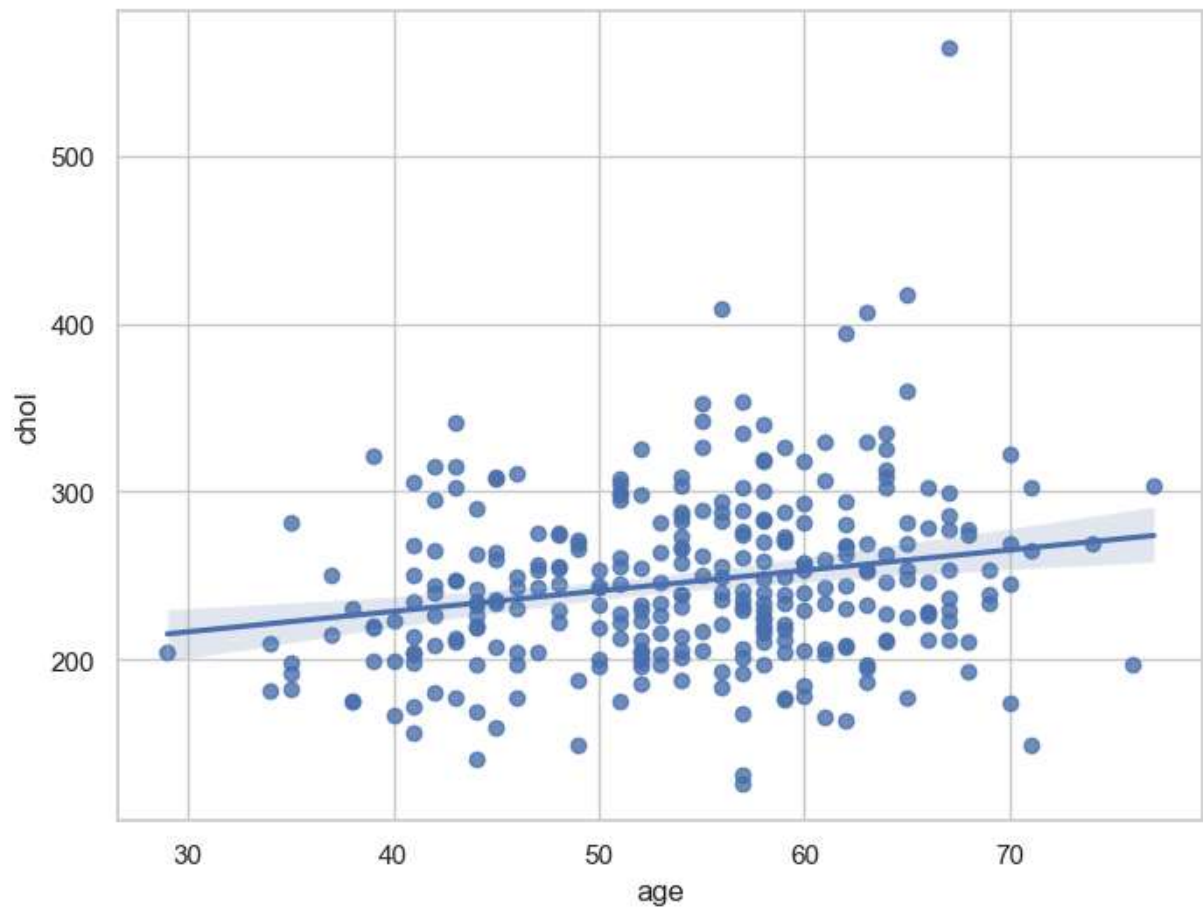
```
In [61]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="age", y="trestbps", data=df)
plt.show()
```

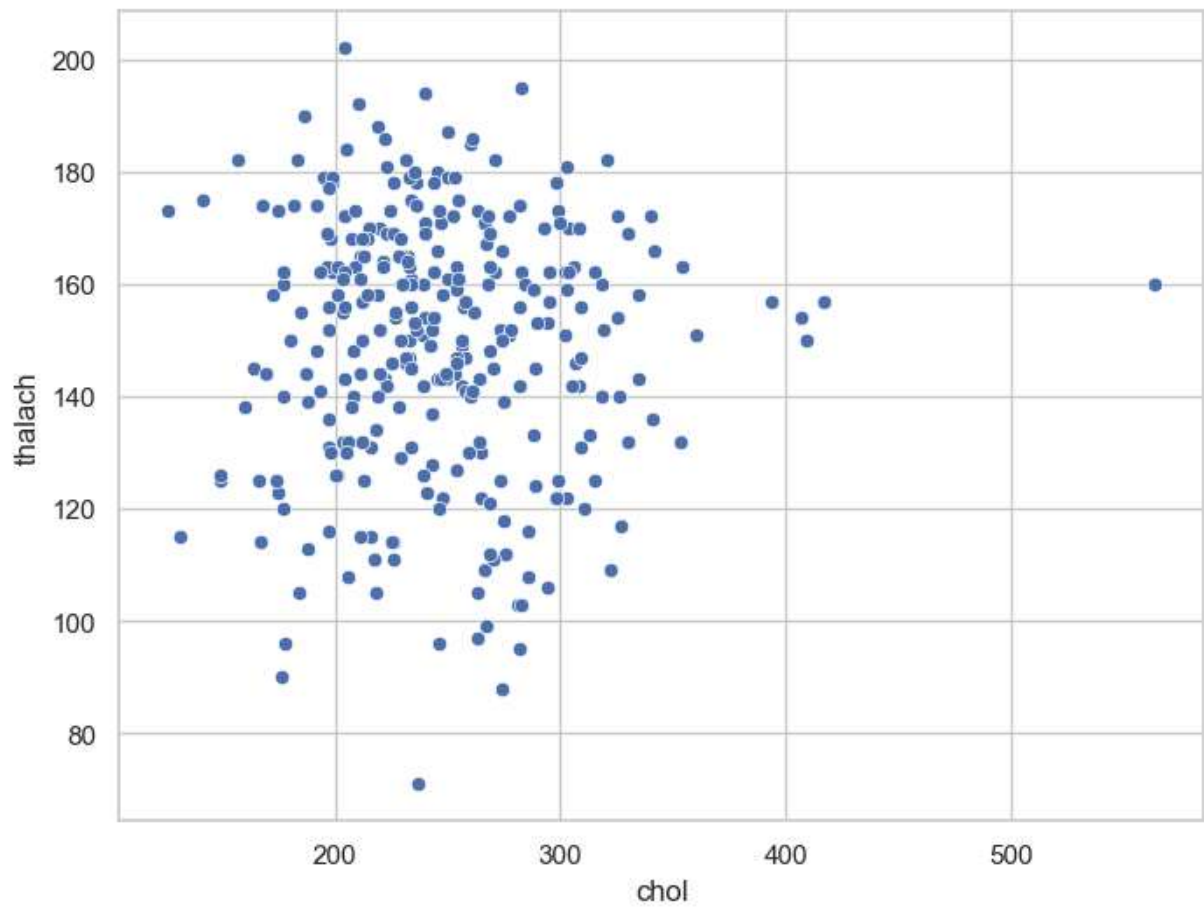
```
In [62]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.scatterplot(x="age", y="chol", data=df)
plt.show()
```



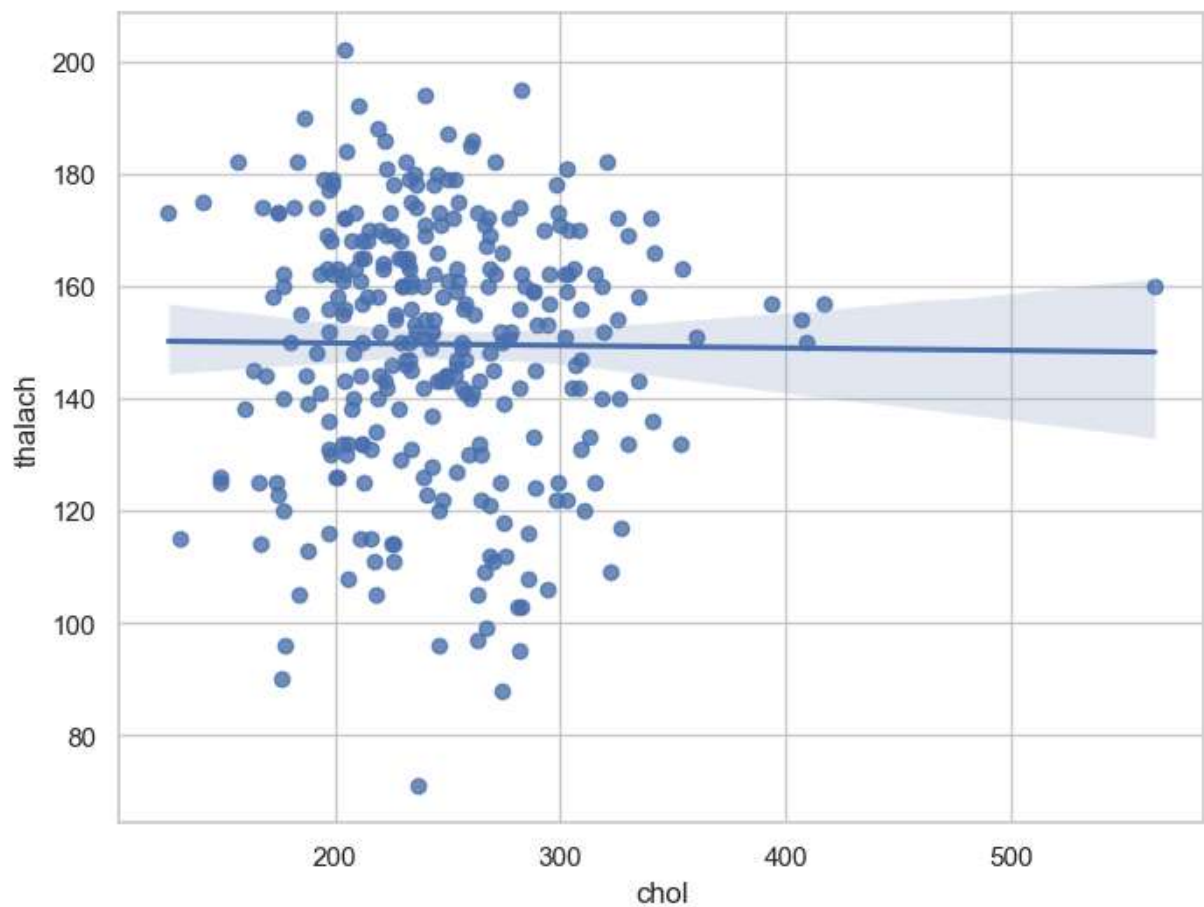
```
In [63]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.regplot(x="age", y="chol", data=df)  
plt.show()
```



```
In [64]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.scatterplot(x="chol", y = "thalach", data=df)
plt.show()
```



```
In [65]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="chol", y="thalach", data=df)
plt.show()
```



```
In [66]: df.isnull().sum()
```

```
Out[66]: age      0
sex        0
cp         0
trestbps   0
chol       0
fbs        0
restecg    0
thalach    0
exang      0
oldpeak    0
slope      0
ca         0
thal       0
target     0
dtype: int64
```

```
In [67]: assert pd.notnull(df).all().all()
```

```
In [68]: assert (df >= 0).all().all()
```

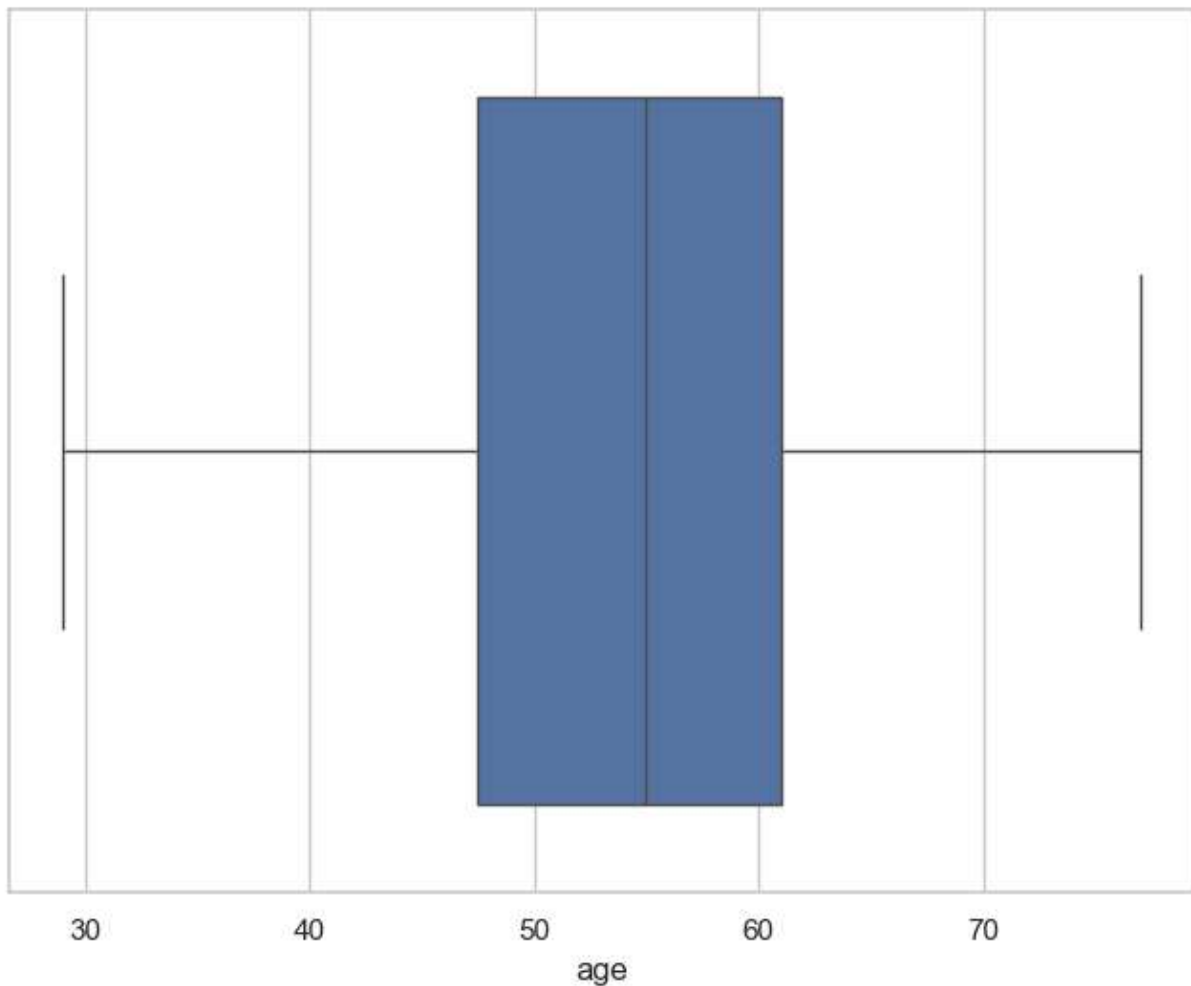
age variable

```
In [79]: df['age'].describe()
```

```
Out[79]: count    303.000000  
mean      54.366337  
std       9.082101  
min       29.000000  
25%      47.500000  
50%      55.000000  
75%      61.000000  
max       77.000000  
Name: age, dtype: float64
```

Box-plot of age variable

```
In [80]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=df["age"])  
plt.show()
```

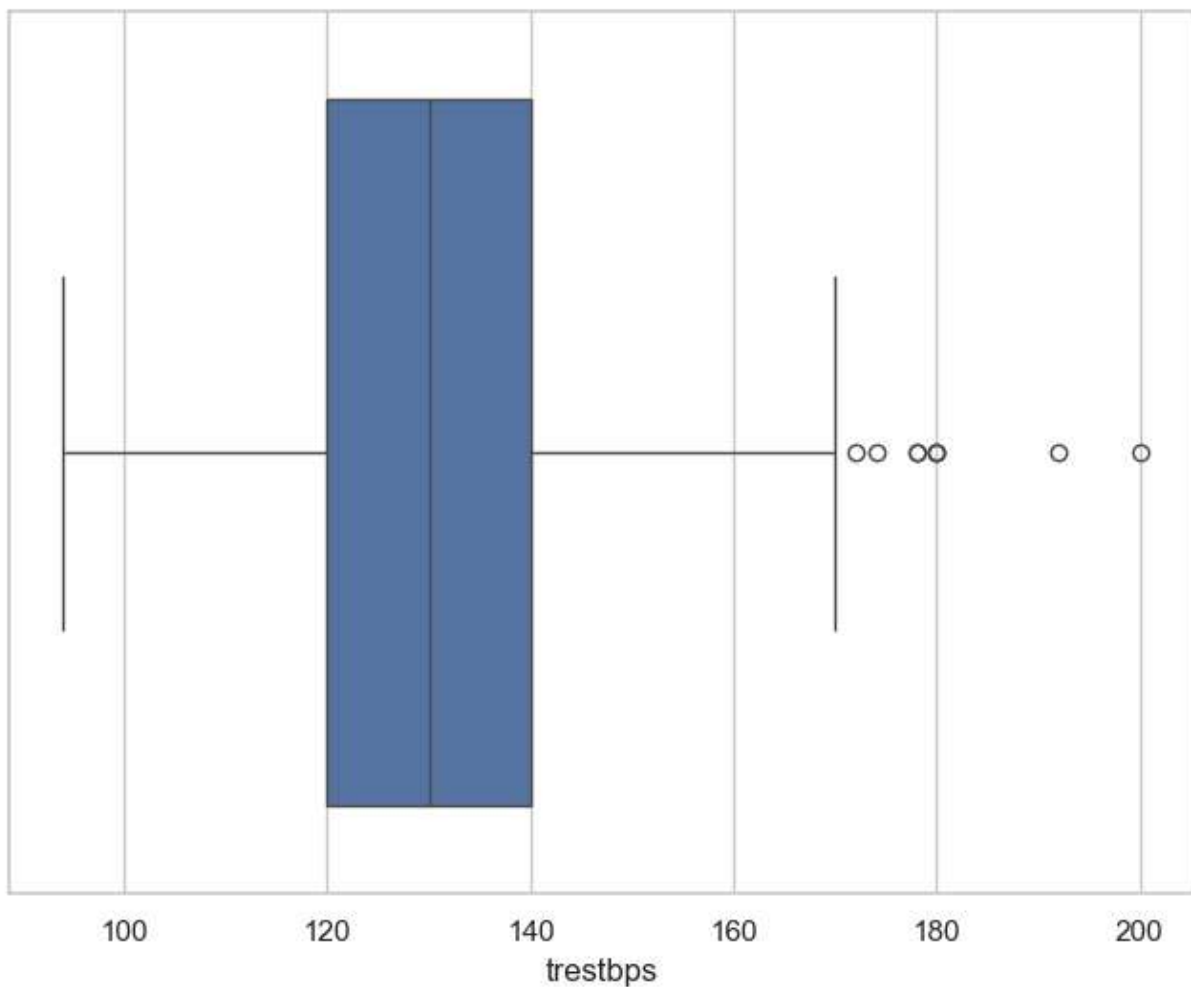


trestbps variable

```
In [81]: df['trestbps'].describe()
```

```
Out[81]: count    303.000000  
mean      131.623762  
std       17.538143  
min       94.000000  
25%      120.000000  
50%      130.000000  
75%      140.000000  
max       200.000000  
Name: trestbps, dtype: float64
```

```
In [82]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=df["trestbps"])  
plt.show()
```



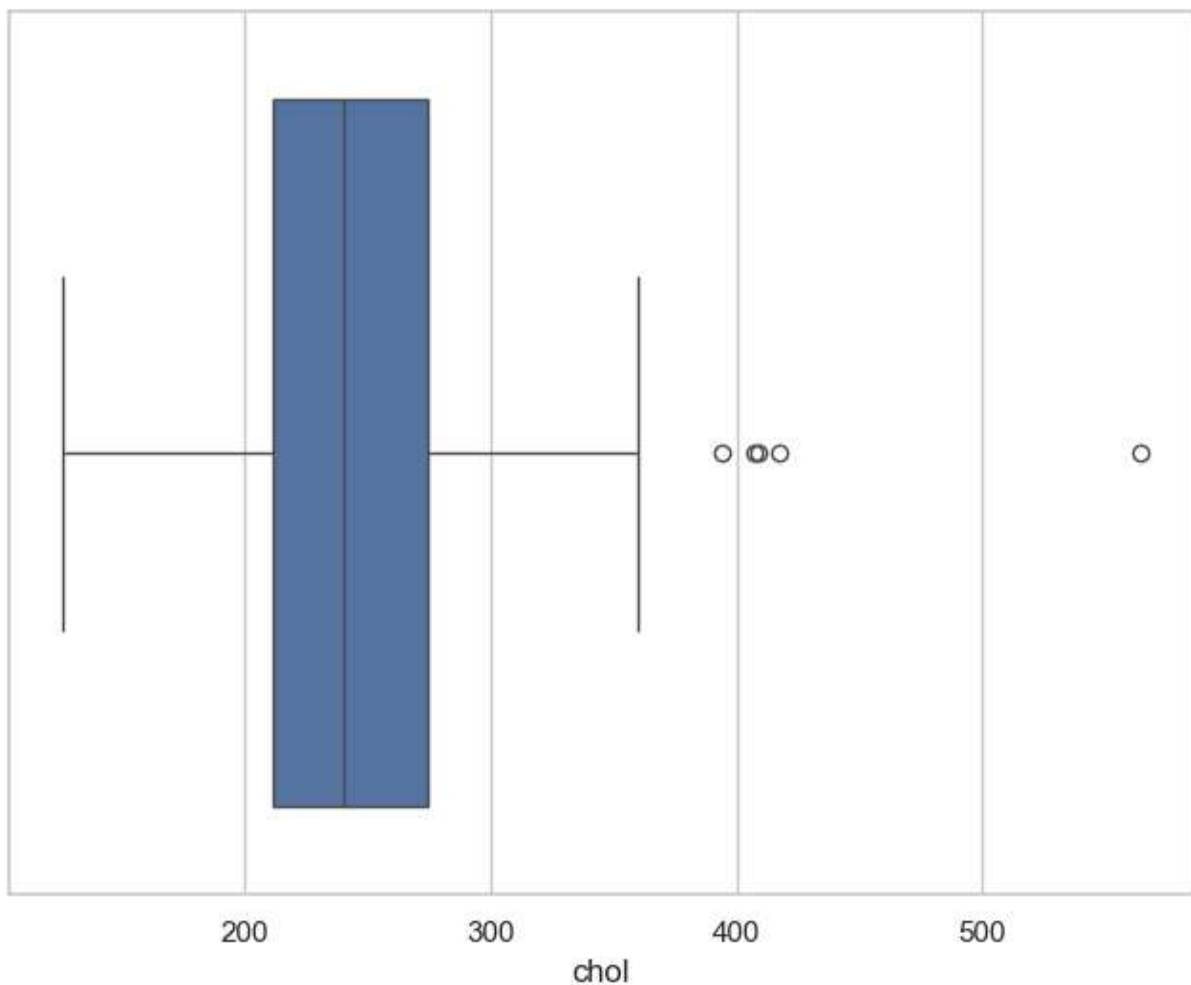
chol variable

```
In [83]: df['chol'].describe()
```

```
Out[83]: count    303.000000  
mean      246.264026  
std       51.830751  
min       126.000000  
25%       211.000000  
50%       240.000000  
75%       274.500000  
max       564.000000  
Name: chol, dtype: float64
```

Box-plot of chol variable

```
In [84]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=df["chol"])  
plt.show()
```



thalach variable

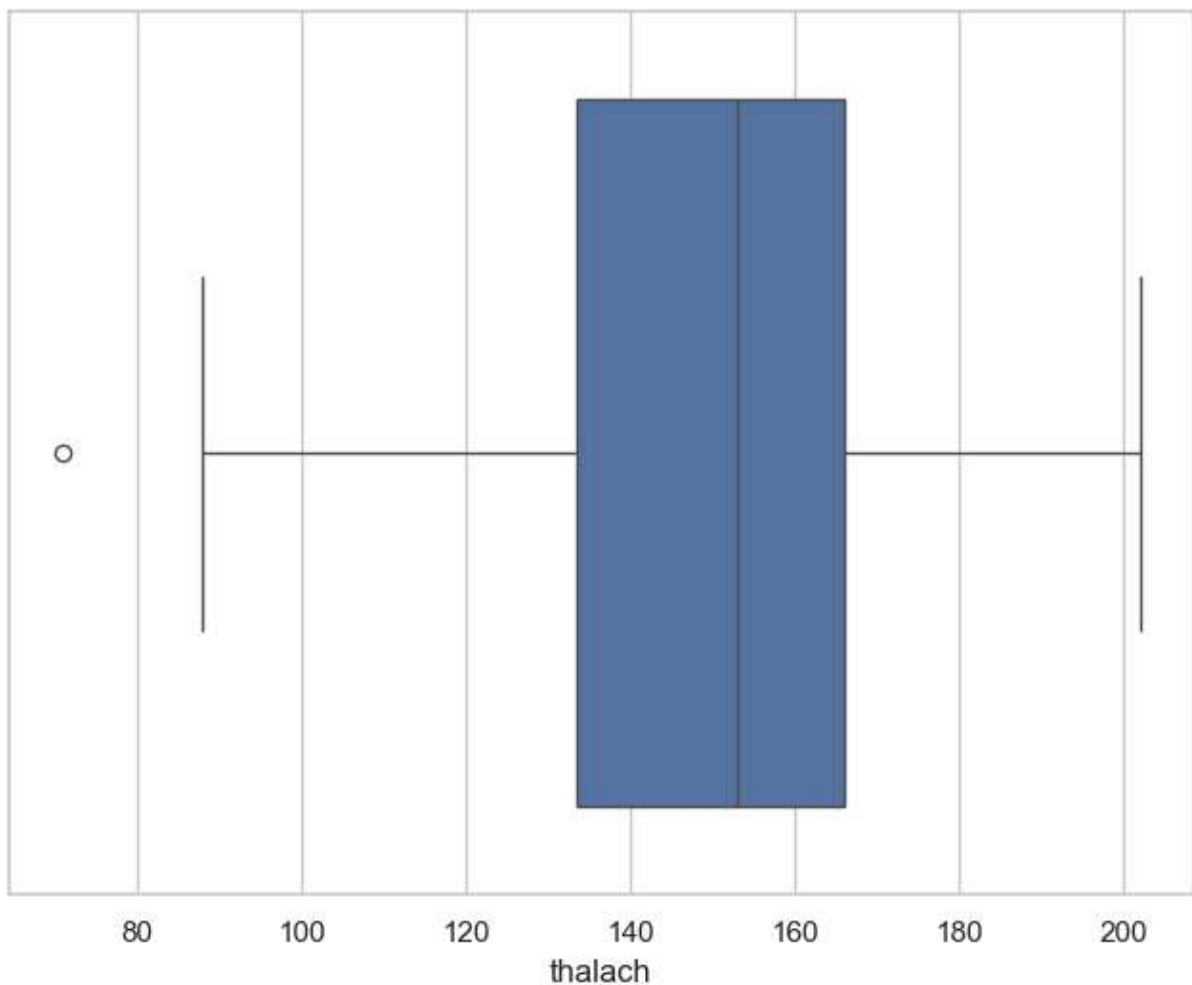
```
In [85]: df['thalach'].describe()
```



```
Out[85]: count    303.000000  
         mean     149.646865  
         std      22.905161  
         min      71.000000  
         25%     133.500000  
         50%     153.000000  
         75%     166.000000  
         max      202.000000  
         Name: thalach, dtype: float64
```

Box-plot of thalach variable

```
In [86]: f, ax = plt.subplots(figsize=(8, 6))  
         sns.boxplot(x=df["thalach"])  
         plt.show()
```



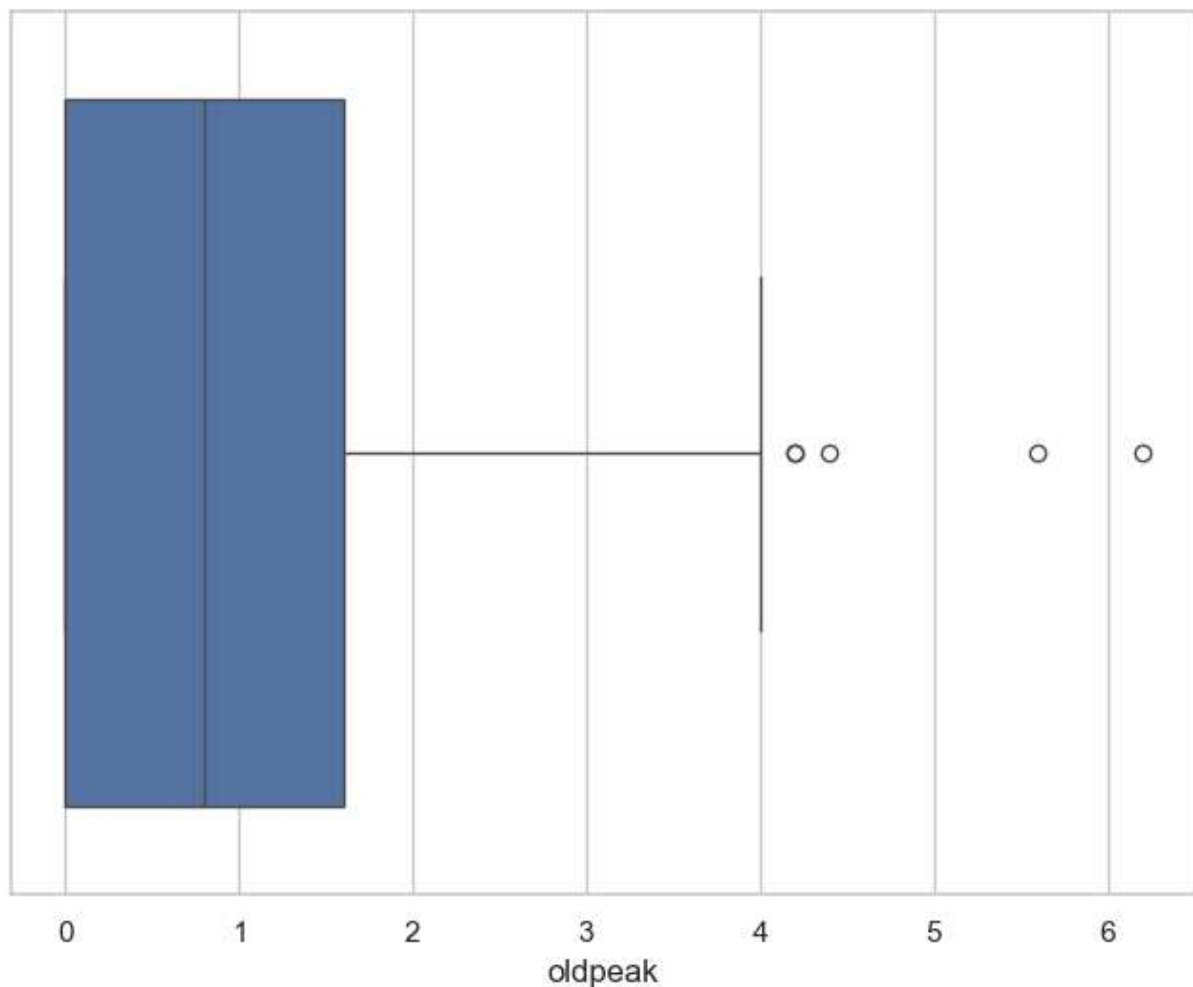
oldpeak variable

```
In [87]: df['oldpeak'].describe()
```

```
Out[87]: count    303.000000  
mean      1.039604  
std       1.161075  
min       0.000000  
25%       0.000000  
50%       0.800000  
75%       1.600000  
max       6.200000  
Name: oldpeak, dtype: float64
```

Box-plot of oldpeak variable

```
In [88]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=df["oldpeak"])  
plt.show()
```



In []:

In []:

In []:

In []:

In []:

In []: